

SUPPLEMENTARY MATERIAL:

Multiobjective Ant Colony Optimization: An Updated Review from Designs to Applications

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SM-1 Introduction

The following tables retain terminology and paper-level fields that are intentionally compressed in the main chapter.

Terminology dictionary

Table [SM-1](#) defines the compact taxonomy labels used in the extended classifications.

SM-2 Benchmark problems – Extended classification

Table [SM-2](#) reports the 53 benchmark profiles under the complete component-oriented classification. † marks milestone studies.

SM-3 Real-world Problems – Extended classification

Table [SM-3](#) retains one row per application study.

Table SM-1: Controlled terminology for the MOACO taxonomical axes and the auxiliary feasibility descriptor used in the extended tables.

Axis	Controlled terms
SE	ACO: Generic ACO engine. ACO_R: Continuous-domain ACO family. ACS: Ant Colony System family. Ant-Q: Ant-Q reinforcement-learning ACO engine. AS: Ant System family. Binary-ACO: ACO variant specialized to binary decision spaces. Mixed: Several ACO engines are explicitly compared. MMAS: MAX-MIN Ant System family. PACO: Population-based ACO. Quantum-ACO: Quantum-inspired ACO engine.
SR	Assignment: Mapping entities to resources, locations, or options. Binary: Fixed binary string/vector where bit position is the solution encoding. Composite: Two or more different representations are mapped to one solution. Configuration: Structured categorical/design configuration. Continuous: Real-valued decision vector. Discrete: Integer/discrete-valued vector not better described as assignment, subset, or permutation. Locus: Locus-based adjacency representation. Path: One ordered path/trajectory through a graph/state space. Permutation: Ordered permutation in which each element appears once. Quantum: Qubit/quantum probability representation. Routes: A set or collection of routes. Sequence: Ordered sequence where values may repeat or are not a permutation. Subset: Unordered selected subset/set of items. Tree: Tree/arborescence solution.
SC	Assignment: Choose an option, resource, or value for an entity. Bitwise: Choose bit values position by position. Decoding: Construct an indirect encoding and decode it into the solution. Levelwise: Choose one discretized level/value for each continuous design variable. Mixed: Several distinct construction rules are coequal in the same study or algorithm. Nodewise: Add the next node, edge, or state incrementally. Positional: Fill an ordered position/stage with a remaining component. Sampling: Sample continuous decision values/distributions. Staged: Build multiple dependent representation layers/stages. Subset-build: Add admissible items to a selected set.
MO	Decomposition: The MOP is decomposed into scalar subproblems/weights. Indicator: An indicator/ranking drives multiobjective selection. Island: Parallel island/migration topology defines the multiobjective strategy. Lexicographic: Objectives are prioritized hierarchically. Mixed: Several multiobjective strategies are coequal. Pareto: Dominance/non-dominated based selection. Preference: Explicit decision-maker/reference/fuzzy preference mechanism. Scalarization: Objectives are combined into one scalar merit value during search.
PM	Shared: A common pheromone structure is used by the search process. Colony-wise: Distinct colonies maintain separate pheromone structures. Objective-wise: Pheromone structures are associated with objectives. Subproblem-wise: Pheromone structures are associated with decomposition subproblems or groups. Component-wise: Different structures encode distinct solution/search components or functions. Variable-wise: Pheromone information is organized by decision variable or variable-level probabilistic model. Mixed: Two or more pheromone-organization schemes are materially combined.
#P	1: One pheromone structure. m: Count scales with the number of objectives. K: Count scales with decomposition groups/subproblems. n: A fixed reported integer greater than one. V: Count varies across configurations or experiments.
HI	Static: A priori heuristic/desirability information is fixed during the run. Dynamic/adaptive: Heuristic information changes with the search state, partial solution, performance, or an adaptive rule. Learned/predicted: A learned model supplies predictions used as heuristic/desirability information. Multi-matrix: Several distinct heuristic structures are maintained and combined or selected. Mixed: Two or more heuristic-information types are materially combined.

Axis	Controlled terms
#H	0 : No heuristic structure. 1 : One heuristic structure. m : Count scales with the objectives. K : Count scales with decomposition groups/subproblems. n : A fixed reported integer greater than one. V : Count varies across configurations or experiments.
CA	Single : One colony/population; no multi-colony topology is present. Objective-wise : Colonies are specialized by objective. Subproblem-wise : Colonies/groups are specialized by decomposition subproblem or region. Heterogeneous : Colonies have different non-objective roles or behaviors. Island : Colonies/islands exchange information through migration. Cooperative : Colonies explicitly share solutions, archives, pheromone information, or search knowledge. Competitive : Colonies explicitly compete. Dynamic : Colony membership, topology, or interaction changes during the run. Mixed : Several materially different colony architectures are compared or combined.
#C	1 : One colony/population. m : Count scales with objectives. K : Count scales with subproblems/groups. n : A fixed reported integer greater than one. V : Count varies across configurations or experiments.
HY	Local-search/trajectory : Local improvement or another trajectory-based intensification augments ACO. Operators/metaheuristic : Operators or search components from another metaheuristic are integrated. Mixed : Two or more augmentation families are materially combined.
LMR	Policy : Learning changes the construction/action policy. Selection : Learning chooses among heuristics, update rules, operators, or algorithmic alternatives. Guidance : Learning produces search guidance. Parameter : Learning controls parameter values. Structure : Learning changes search organization or structural components. Design : Learning is used to design/configure the algorithm. Mixed : Learning serves more than one material role.
LMM	RL : Other reinforcement-learning method. GNN : graph neural network guidance. GPT-4 : language-model guidance Other/open : Other learning models.
PA	Adaptive : Parameters change by an explicit adaptive rule not better classified below. Scheduled : Parameters follow a time, iteration, or evaluation schedule. Learned : A learning mechanism controls parameters or operator choices. Feedback : Parameters respond to measured search state or performance. Mixed : Two or more adaptation modes are materially combined.
FP	Decoder : An indirect representation is decoded into a feasible evaluated solution. Filter : Infeasible moves/components are removed before construction proceeds. Hybrid : Two feasibility-preserving mechanisms are combined. Intrinsic : The representation/construction domain itself can only express feasible candidates for the relevant constraints. Repair : A constructed candidate is repaired before full evaluation/acceptance.

Table SM-2: Extended MOACO benchmark studies under the component-oriented taxonomy adopted in the main chapter. **#O** = number of objectives; **Size** = reported instance or problem scale; **Eq./Ineq.** = equality/inequality constraints; **SE** = search engine; **SR** = solution representation; **SC** = solution construction; **MO** = multiobjective coordination; **PM/#P** = pheromone organization / number of pheromone structures; **HI/#H** = heuristic-information organization / number of heuristic structures; **CA/#C** = colony architecture / number of colonies; **HY** = hybridization; **LMR/LMM** = learning role / learning method; **PA** = parameter adaptation; and **FP** = feasibility preservation. The corresponding full terms are defined in Table SM-1. A superscript † identifies studies classified as milestones in Table ??.

Study	Benchmark problem /instances	#O	Size	Eq./Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
1999 MACS-VRPTW ^{1†}	Vehicle routing problem with time windows (VRPTW) /Solomon C1/C2/R1/R2/RC1/RC2 benchmark instances	2	100 customers	Y /Y	ACS	Routes	Nodewise	Lexicographic	Obj.-wise /m	Static /1	Obj.-wise + Cooperative /m	Local-search /trajectory	-	-	Filter
1999 MOAQ ^{2†}	Artificial binary test problem /Instances with 12 and 28 bits	2	12/28 bits	N /N	Ant-Q	Binary	Bitwise	Pareto	Obj.-wise /m	Multi-matrix /2	Obj.-wise + Cooperative /m	-	Policy /Q-learning	-	Intrinsic

Table SM-2: Extended MOACO benchmark studies — continued.

Study	Benchmark problem /instances	#O	Size	Eq. /Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
1999 MOAQ ^{2†}	Schaffer F2 /Instances encoding 14-bit	2	1 var. / 14-bit encoding	N /N	Ant-Q	Binary	Bitwise	Pareto	Obj.-wise /m	- /0	Obj.-wise + Coopera- tive /m	-	Policy /Q-learning	-	Intrinsic
2001 MOACO ^{3†}	Single-machine scheduling /Instances with random 100-job	2	100 jobs	N /N	ACS	Permuta- tion	Positional	Pareto	Obj.-wise /m	Dynamic /adaptive /1	Obj.-wise + Coopera- tive /m	-	-	-	-
2002 SACO ⁴	Flow shop scheduling /Random instances; 10 sizes, 50 per size	2	10–200 jobs	N /N	MMAS	Permuta- tion	Positional	Lexico- graphic	Shared /1	- /0	Single /1	Local- search /trajectory	-	Sched- uled	Filter
2003 MONACO ⁵	Shortest path problem /2 test networks: 18 nodes/27 arcs and 25 nodes/40 arcs	2	18/25 nodes	N /N	AS	Path	Nodewise	Scalariza- tion	Shared /1	Static /1	Single /1	-	-	-	-
2003 PACO ^{6†}	Single-machine scheduling /Generated tardiness/setup- cost cases; 2 biobjective + 1 four-criterion instance	2,4	100 jobs	N /N	PACO	Permuta- tion	Positional	Pareto	Obj.-wise /m	Dynamic /adaptive /1	Single /1	-	-	Feed- back	Intrinsic
2004 MOACO ⁷	Quadratic assignment problem /6 bQAP instances, n=50; Knowles–Corne generator, 3 correlation levels	2	50 facilities	Y /N	MMAS	Assign- ment	Assign- ment	Scalariza- tion	Shared /1	Static /1	Mixed /V	Local- search /trajectory	-	-	Filter
2007 m-ACO ^{8†}	Knapsack problem /MOKP set: 100/250/500 items	2	100/250 500 items	N /Y	Mixed ^a	Subset	Subset- build	Pareto	Mixed /V	Multi- matrix /V	Mixed /V	-	-	-	Filter
2007 CPACO ⁹	Traveling salesman problem (TSP) /TSPLIB-derived bTSP: 100/150/200 cities + one 4-objective TSP	2,4	100/150 200 cities	N /N	PACO	Permuta- tion	Nodewise	Pareto	Shared /1	Multi- matrix /m	Single /1	-	-	-	Filter
2007 PACO-MOFO ^{10†}	Continuous multiobjective function optimization /MOP1–MOP3 and MOP6	2	2 vars.	N /N	PACO	Continu- ous	Sampling	Pareto	Variable- wise /1	- /0	Single /1	-	-	Sched- uled	-
2007 MOACO _R ^{11†}	Continuous multiobjective function optimization /SCH1–2; DEB1–2; POL; FON; KUR; ZDT1–4; ZDT6	2	1–30 vars.	N /N	ACO _R	Continu- ous	Sampling	Pareto	Variable- wise /1	- /0	Single /1	-	-	-	-
2007 ACO-LQDF ^{12†}	Flexible job shop scheduling /15 jobs, 56 operations, 10 machines	3	15 jobs / 56 ops. / 10 machines	N /N	MMAS	Assign- ment	Assign- ment	Scalariza- tion	Comp.-wise /2	Static /1	Single /1	Local- search /trajectory	-	-	Filter
2007 MOACO ¹³	Flow shop scheduling /Reeves: 21 instances (20–75 jobs; 5–20 machines)	3	20–75 jobs	N /N	ACS	Permuta- tion	Positional	Lexico- graphic	Shared /1	Static /1	Single /1	Local- search /trajectory	-	-	Filter
2008 AMPACOA ¹⁴	Biobjective traveling salesman problem (bTSP) /kroAB100, kroAC100, kroCB100, kroAB150, kroAB200	2	100/150 200 cities	N /N	PACO	Permuta- tion	Nodewise	Pareto	Obj.-wise /m	Multi- matrix /m	Single /1	-	-	Sched- uled	Filter

Table SM-2: Extended MOACO benchmark studies — continued.

Study	Benchmark problem /instances	#O	Size	Eq./Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2008 LAH ¹⁵	Shortest path problem /8 exactly solvable test instances incl. D1	3	$ V + E $ 527–1408	N /N	Mixed ^b	Path	Nodewise	Pareto	Shared /1	Mixed /V	Coopera- tive /V	-	-	-	Filter
2010 MOACSA ¹⁶	Flow shop scheduling /Taillard: 20/50/100-job instances	2	20/50/100 jobs	N /N	ACS	Permuta- tion	Positional	Scalariza- tion	Shared /1	Static /1	Single /1	Local- search /trajectory	-	-	Filter
2011 HACO ¹⁷	Open shop scheduling /Taillard: 9 instances + randomly generated due dates	2	4×4 5×5 7×7 jobs $\times$ machines	N /Y	ACO	Permuta- tion	Decoding	Pareto	Shared /1	Static /1	Single /1	Mixed	-	Feed- back	Decoder
2012 MoACO/D ^{18†}	Traveling salesman problem (TSP) /10 bTSP instances: KroAB50, KroCD50, KroAB100, KroAD100, KroBC100, KroCD100, KroAB150, KroAB200, EuclidAB300, EuclidCD300	2	50/100 150/200 300 cities	N /N	MOEA/D	Permuta- tion	Nodewise	Decompo- sition	Subprob.- wise /K	Multi- matrix /K	Subprob.- wise + Coopera- tive /K	-	-	-	Filter
2012 MOACO ¹⁹	Traveling salesman problem (TSP) /108 training instances (100/200/300 cities) + 12 test instances (100/200/300/500 cities)	2	100/200 300/500 cities	N /N	Mixed ^c	Permuta- tion	Nodewise	Scalariza- tion	Mixed /V	Multi- matrix /V	Mixed /V	Local- search /trajectory	-	-	Filter
2013 MOEA/D- ACO ^{20†}	Knapsack problem /9 MOKP instances with 250/500/750 items	2–4	250/500 750 items	N /Y	MOEA/D	Subset	Subset- build	Decompo- sition	Subprob.- wise /K	Static /1	Subprob.- wise + Coopera- tive /K	-	-	-	Filter
2013 MOEA/D- ACO ^{20†}	Traveling salesman problem (TSP) /12 MTSP instances with 100/200/300 cities	2	100/200 300 cities	N /N	MOEA/D	Permuta- tion	Nodewise	Decompo- sition	Subprob.- wise /K	Static /1	Subprob.- wise + Coopera- tive /K	-	-	-	Filter
2013 iBIANT / iMOACS / iCHAC ^{21†}	Traveling salesman problem (TSP) /TSPLIB: kroAB100, kroAB150, kroAB200	2	100/150 200 cities	N /N	Mixed ^d	Permuta- tion	Nodewise	Pareto	Mixed /V	Multi- matrix /m	Island /V	Local- search /trajectory	-	Feed- back	Filter
2013 MOACO ²²	Quadratic assignment problem /KC BiQAP instances + random n=30/40/50	2	30/40/50 facilities	Y /N	Mixed ^e	Assign- ment	Assign- ment	Pareto	Mixed /V	Multi- matrix /V	Mixed /V	-	Policy /Q-learning	Feed- back	Filter
2015 MOACO ²³	Traveling salesman problem (TSP) /6 reference TSP instances (50 and 100 cities)	2,3,4	50/100 cities	N /N	Mixed ^f	Permuta- tion	Nodewise	Pareto	Mixed /V	Multi- matrix /V	Mixed /V	-	-	Mixed ^f	Filter
2015 IBACO ^{24†}	Multidimensional knapsack problem /9 MKP benchmarks with 250/500/750 items	2,3,4	250/500 750 items	N /Y	AS	Subset	Subset- build	Indicator	Shared /1	Static /1	Single /1	-	-	-	Filter

Table SM-2: Extended MOACO benchmark studies — continued.

Study	Benchmark problem /instances	#O	Size	Eq./Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2016 iPM-MOACOs ²⁵	Traveling salesman problem (TSP) /4 primary benchmark instances (100/150/200 cities) + 1 hypervolume case	2	100/150 200 cities	N /N	Mixed ^g	Permutation	Nodewise	Pareto	Mixed /V	Multi-matrix /V	Single /1	-	-	Feed-back	Filter
2017 iMOACO _R ^{26†}	Continuous multiobjective benchmark problems /ZDT, DTLZ and WFG suites; DTLZ2 scalability study	2–10	10–30 vars.	N /N	ACO _R	Continuous	Sampling	Indicator	Variable-wise /1	- /0	Single /1	-	-	-	-
2017 MACO ²⁷	Assembly line balancing /Scholl reference instances (Mitchell–Arcus2)	3	21–111 tasks	Y /Y	ACO	Assignment	Assignment	Lexicographic	Shared /1	Static /1	Single /1	-	-	-	Filter
2018 MOACO ²⁸	Multiobjective continuous benchmark problems /ZDT1–ZDT3	2	30 vars.	N /N	ACO	Continuous	Sampling	Scalarization	Shared /1	- /0	Single /1	-	-	-	-
2019 MATA/ERU ^{29†}	Multidimensional knapsack problem /12 ETH benchmarks with 100/250/500/750 items	2,3,4	100/250 500/750 items	N /N	AS	Subset	Subset-build	Indicator	Comp.-wise /2	Static /1	Single /1	-	-	-	-
2019 QACMOR ^{30†}	Wireless sensor network routing /WSN simulations + physical steel-plant deployment	3	10–100 nodes	N /N	Quantum-ACO	Quantum	Assignment	Scalarization	Shared /1	Static /1	Single /1	Operators Quantum EAs	-	Feed-back	Filter
2019 Gw-ACO ³¹	Multidimensional knapsack problem /9 MKP instances (250/500/750 items)	2,3,4	250/500 750 items	N /Y	ACO	Subset	Subset-build	Pareto	Shared /1	Multi-matrix /m	Single /1	-	-	Scheduled	Filter
2019 MOACO ³²	Traveling salesman problem (TSP) /9 TSP benchmarks incl. kroAB/AC/AD/CD/CE100, euclidAB/CE100, kroAB150/200	2	100/150 200 cities	N /N	Mixed ^h	Permutation	Nodewise	Mixed ^h	Mixed /V	Static /1	Mixed /V	-	-	Feed-back	Filter
2020 RL-MOACO ^{33†}	Resource allocation /Simulated MEC resource-allocation scenario	-	550 nodes	N /N	ACO	Assignment	Assignment	Pareto	- /-	Static /1	- /-	Local-search /trajectory	Policy /RL	-	-
2020 MOCD-ACO ^{34†}	Community detection / graph partitioning /Synthetic networks + four real networks	2	34/62 105/115 nodes (real)	N /N	MOEA/D	Locus	Decoding	Decomposition	Subprob.-wise /K	Static /1	Subprob.-wise + Cooperative /K	Mixed	-	Feed-back	Decoder
2020 NSACO-GA ³⁵	Resource-constrained project scheduling /PSPLIB/ProGen RCPSP instances	3	10–30 activities	Y /Y	ACO	Composite	Decoding	Pareto	Comp.-wise /2	Multi-matrix /2	Single /1	Mixed	-	-	Decoder
2020 MaOACO/D-RP ³⁶	Traveling salesman problem (TSP) /TSPLIB-derived 100-node multiobjective TSP instances	2,5 10	100 nodes	N /N	MOEA/D	Permutation	Nodewise	Decomposition	Subprob.-wise /K	Static /1	Subprob.-wise + Cooperative /K	-	-	Feed-back	Filter

Table SM-2: Extended MOACO benchmark studies — continued.

Study	Benchmark problem /instances	#O	Size	Eq./Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2022 RLMOEA/D-ACO ^{37†}	Traveling salesman problem (TSP) /9 bTSP instances: kroAB/AC/AD/CD/CE100, euclidAB/CE100, kroAB150/200	2	100/150 200 cities	N /N	MOEA/D	Permutation	Nodewise	Decomposition	Subprob.-wise /K	Static /1	Subprob.-wise + Cooperative /K	-	Selection /Learning automata	Learned	Filter
2022 MaOACO/D-DPS ^{38†}	Traveling salesman problem (TSP) /9 TSPLIB-derived cases: 3 bi-, 3 five-, and 3 ten-objective	2,5 10	100 nodes	N /N	MOEA/D	Permutation	Nodewise	Decomposition	Subprob.-wise /K	Static /1	Subprob.-wise + Cooperative /K	-	Selection /SARSA	Learned	Filter
2023 INSA ^{39†}	Feature selection /13 classification datasets, including Madelon	2	18–5147 features 72–2600 samples	N /N	Binary-ACO	Subset	Subset-build	Pareto	Shared /1	Static /1	Single /1	-	-	-	Filter
2024 ϵ -MOACO ^{40†}	Multiobjective continuous benchmark problems /12 problems from ZDT/BNH/SRN/TNK/DTLZ	2–3	2–30 vars.	N /Y	ACO _R	Continuous	Sampling	Pareto	Variable-wise /1	- /0	Single /1	Mixed	-	Feed-back	-
2024 GI-MOACO _R ^{41†}	Continuous multiobjective benchmark problems /DTLZ/WFG + inverted variants; IMOP/VIE	2,3 5,7	6–34 vars. (DTLZ/WFG)	N /N	ACO _R	Continuous	Sampling	Pareto	Variable-wise /1	- /0	Single /1	-	-	Scheduled	-
2025 IACO-QL ^{42†}	Shortest path problem /Arctic route planning; 2 representative + 10 additional ice-chart dates	3	12 dates 10 km × 10 km grid cell	N /Y	ACO	Path	Nodewise	Scalarization	Shared /1	Static /1	Single /1	-	Policy /Q-learning	Feed-back	Filter
2025 D-ACO ^{43†}	Shortest path problem /10 × 10 × 10 3-D grid with random obstacles	3	10 × 10 × 10 grid	N /N	ACO	Path	Nodewise	Scalarization	Shared /1	Mixed /2	Single /1	-	Policy /DQN	Feed-back	Filter
2025 mACO/OL ^{44†}	Vehicle routing problem (VRP) /30 synthetic HRSP-DOA instances + steel-company industrial case	2	120–210 initial orders +15–30 per event 300 initial (real)	Y /Y	ACO	Routes	Nodewise	Indicator	Shared /1	Mixed /V	Single /1	-	Selection /Bandit	Learned	Filter
2025 GNN-wACO ^{45†}	Multidimensional knapsack problem /12 MOMKP instances; 6 learning + 6 comparative	2,3,4	100/250 500/750 items	N /Y	ACO	Subset	Subset-build	Pareto	Shared /1	Mixed /2	Single /1	-	- /GNN	Scheduled	Filter
2025 MOACO-DCE ^{46†}	Constrained multiobjective continuous optimization /MW and CF benchmark suites	2–3	10/15 vars.	N /Y	ACO	Continuous	Sampling	Pareto	Shared /1	- /0	Single /1	Local-search /trajectory	-	Feed-back	Repair
2026 MACO/NDS ⁴⁷	Knapsack problem /3 MKP benchmarks (30/40/50 items)	4–6	30/40/50 items	Y /Y	ACO	Subset	Subset-build	Pareto	Mixed /V	Multi-matrix /V	Single /1	-	-	-	Filter
2026 MMOAC-ENCEI ⁴⁸	Multimodal multiobjective optimization /MMF1–MMF6	2	2 vars.	N /N	ACO	Continuous	Sampling	Pareto	Shared /1	- /0	Single /1	-	-	Feed-back	-

Table SM-2: Extended MOACO benchmark studies — continued.

Study	Benchmark problem /instances	#O	Size	Eq./Ineq.	SE	SR	SC	MO	PM/#P	HI/#H	CA/#C	HY	LMR/LMM	PA	FP
2026 FRU-GNN-wACO ^{49†}	Multidimensional knapsack problem /9 Zitzler–Thiele-derived MOMKP instances	2,3,4	250/500 750 items	N /Y	ACO	Subset	Subset-build	Pareto	Shared /1	Static /1	Single /1	-	- /GPT-4	Scheduled	Filter
2026 MOACO-DQN ^{50†}	Cluster head selection /MATLAB WSN deployments with 50–200 random nodes	3	50–200 nodes	N /N	ACO	Subset	Assignment	Scalarization	Shared /1	Static /1	Single /1	-	Policy /DQN	-	-
2026 GNN-ACO ^{51†}	Flexible job shop scheduling /Realistic cable-production benchmarks with disruptions	4	10–150 orders 100–2700 ops.	Y /Y	ACO	Assignment	Assignment	Pareto	Shared /1	Mixed /2	Single /1	-	- /GNN	Feedback	Filter

Mixed-category notes. ^a SE compares m-ACO1–m-ACO4; their pheromone and colony organizations also differ. ^b SE combines Ant System and Ant Colony System. ^c SE compares BicriterionAnt, MACS, COMPETants, P-ACO, and m-ACO1–4 on a common MMAS implementation; PM and CA therefore vary by configuration. ^d SE compares Biant, MOACS, and CHAC; the distributed variants use an island topology. ^e SE compares BA/BA-C, UBMC/UBMC-C, PBMC/PBMC-C, CompetAnts, MONACO, MACS, PACO/PACO-C, and MOAQ. ^f SE compares ACOMOFS, AMPACOA, CPACO, mACO4, MACS, MCAA, and PSACO; PA is algorithm-specific. ^g SE applies the iPM strategy to PACO, Biant, and MACS. ^h SE compares several MOACO architectures; their pheromone and colony organizations are consequently classified as mixed.

Table SM-3: Extended MOACO application classification under the component-oriented taxonomy adopted in the main chapter. **#O** = number of objectives; **Size** = reported instance or scenario scale; **Eq./Ineq.** = equality/inequality constraints; **SE** = search engine; **SR** = solution representation; **SC** = solution construction; **MO** = multiobjective coordination; **PM/#P** = pheromone organization / number of pheromone structures; **HI/#H** = heuristic-information organization / number of heuristic structures; **CA/#C** = colony architecture / number of colonies; **HY** = hybridization; **LMR/LMM** = learning role / learning method; **PA** = parameter adaptation; and **FP** = feasibility preservation. The corresponding full terms are defined in Table SM-1.

Year	Domain	Canonical problem /Specific application	#O	Size	Eq./Ineq.	SE	SR	SC	MO	PM/#P	HI/#H	CA/#C	HY	LMR/LMM	PA	FP
1 — Routing, tours and paths																
1999	Transportation	Vehicle routing problem (VRP) ¹ /Solomon VRPTW: 56 C1/C2/R1/R2/RC1/RC2 instances	2	100 customers	Y /Y	ACS	Routes	Nodewise	Lexicographic	Obj.-wise /m	Static /1	Obj.-wise + Cooperative /m	Local-search /trajectory	-	-	Filter
2003	Transportation	Vehicle routing problem (VRP) ⁵² /Solomon VRPTW instances (including C1/RC2)	3	100 customers	Y /Y	ACS	Routes	Nodewise	Pareto	Shared /1	Multi-matrix /2	Single /1	-	-	Feedback	Filter
2005	Communications	Multicast routing problem ⁵³ /Japan NTT topology (55 nodes, 144 links)	4	55 nodes / 144 links	N /Y	Mixed	Tree	Nodewise	Pareto	Shared /1	Multi-matrix /3	Single /1	-	-	-	Filter
2006	Defense	Shortest path problem ⁵⁴ /two 45×45 Panzer General maps	2	45×45 grid	N /Y	Mixed	Path	Nodewise	Preference	Obj.-wise /m	Static /1	Single /1	-	-	Feedback	Filter
2009	Transportation	Pickup and delivery problem with time windows ⁵⁵ /preliminary real data + SOG-generated realistic instances	2	20/60/100 switch orders	Y /Y	AS	Routes	Assignment	Scalarization	Mixed /6	Static /1	Heterogeneous + Dynamic /2	-	-	Feedback	Filter
2011	Communications	Shortest path problem ⁵⁶ /NS-2.34 mobility/network scenarios	3	20–100 nodes	N /N	ACO	Path	Nodewise	Preference	Shared /1	Static /1	Single /1	-	-	-	-

Table SM-3: Extended MOACO application classification — continued.

Year	Domain	Canonical problem /Specific application	#O	Size	Eq. /Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2011	Transportation	Shortest path problem ⁵⁷ /Wuhan Sports Center network; 25,000 simulated evacuees	3	25,000 evacuees	Y /Y	Binary-ACO	Routes	Assignment	Lexicographic	Shared /1	Static /1	Single /1	-	-	-	Filter
2013	Engineering	Shortest path problem ⁵⁸ /ship Track-Keeping Control System diagnosis	3	26 fault nodes	N /Y	ACO	Path	Nodewise	Pareto	Shared /1	- /0	Single /1	-	-	-	Filter
2016	Transportation	Vehicle routing problem (VRP) ⁵⁹ /33 adapted literature benchmarks with added time/energy criteria	3	33 benchmark instances	Y /Y	PACO	Routes	Nodewise	Preference	Obj.-wise /3	Static /1	Single /1	-	-	-	Filter
2016	Transportation	Shortest path problem ⁶⁰ /Tehran transport-network data	3	-	N /N	PACO	Path	Nodewise	Pareto	Obj.-wise /m	Static /1	Mixed /V	-	-	Adaptive	-
2018	Engineering	Traveling salesman problem (TSP) ⁶¹ /TSPLIB cases + randomly generated scenarios	2	30–300 targets 10–20 robots	Y /Y	MMAS	Assignment	Assignment	Decomposition	Subprob.-wise /K	Static /1	Single /1	-	-	-	Filter
2018	Transportation	Vehicle routing problem (VRP) ⁶² /randomly generated small and large instances	2	10–200 stations 156–2348 students	Y /Y	ACO	Routes	Assignment	Pareto	Mixed /4	Static /1	Single /1	Local-search /trajectory	-	-	Hybrid
2019	Transportation	Vehicle routing problem (VRP) ⁶³ /six small random instances + 10 large simulated networks	4	6 small + 10 large networks	Y /Y	AS	Routes	Nodewise	Scalarization	Shared /1	Static /1	Single /1	-	-	-	Filter
2019	Manufacturing	Wireless sensor network routing ³⁰ /WSN simulations + physical steel-plant deployment	3	10–100 nodes	N /N	Quantum-ACO	Quantum	Assignment	Scalarization	Shared /1	Static /1	Single /1	-	-	Feed-back	Filter
2019	Communications	Wireless sensor network routing ⁶⁴ /NS-2, 100–500 nodes	2	100–500 nodes	N /Y	ACO	Path	Nodewise	Pareto	Shared /1	Multi-matrix /3	- /-	Operators /meta-heuristic	-	-	Filter
2019	Communications	Wireless sensor network routing ⁶⁵ /NS-2, 100 random nodes with simulated attacks	2	100 nodes	N /Y	ACO	Path	Nodewise	Pareto	Obj.-wise /m	Multi-matrix /2	Single /1	Operators /meta-heuristic	-	-	Filter
2019	Transportation	Vehicle routing problem (VRP) ⁶⁶ /modified Solomon benchmarks + Chinese logistics-company case	2	Solomon benchmarks + 1 real case	Y /Y	ACO	Routes	Nodewise	Scalarization	Shared /1	Static /1	Single /1	-	-	-	Filter
2020	Communications	Multicast routing problem ⁶⁷ /three generated multicast networks	3	3 networks first = 10 nodes	N /N	AS	Tree	Nodewise	Scalarization	Shared /1	Static /1	- /-	-	-	-	-
2020	Transportation	Vehicle routing problem (VRP) ⁶⁸ /CVRP benchmark instances	2	51–200 customers	N /N	ACS	Routes	Nodewise	Pareto	Obj.-wise /2	Multi-matrix /3	Single /1	Mixed ^a	-	-	Filter
2020	Communications	Cognitive-radio network routing ⁶⁹ /random-node cognitive-radio/WSN scenarios	-	50–350 nodes	N /N	ACO	Path	Nodewise	Pareto	Shared /1	Static /1	Single /1	Operators /meta-heuristic	Policy /Double Q-learning	-	-

Table SM-3: Extended MOACO application classification — continued.

Year	Domain	Canonical problem /Specific application	#O	Size	Eq. /Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2020	Transportation	Vehicle routing problem (VRP) ⁷⁰ /Solomon VRPTW instances	3	100 customers	Y /Y	ACS	Routes	Nodewise	Pareto	Shared /1	Static /1	Single /1	-	-	-	Filter
2021	Transportation	Vehicle routing problem (VRP) ⁷¹ /Solomon VRPTW instances	3	100 customers	Y /Y	ACS	Routes	Nodewise	Pareto	Obj.-wise /m	Static /1	Cooperative /3	Local-search /trajectory	-	-	-
2021	Transportation	Shortest path problem ⁷² /MS Smyril operations + ECMWF weather data	2	41×81 grid nodes	N /Y	ACO	Path	Nodewise	Pareto	Shared /1	Dynamic /adaptive /1	Single /1	Operators /meta-heuristic	-	Feed-back	Filter
2021	Communications	Multicast routing problem ⁷³ /10-node MANET multicast simulation	3	10 nodes	N /N	-	Tree	Nodewise	Scalarization	Shared /1	Static /1	- /-	-	-	-	-
2021	Communications	Wireless sensor network routing ⁷⁴ /10/20/40-node WSN scenarios	2	10/20/40 nodes	N /N	ACO	Routes	Nodewise	Scalarization	Shared /1	Static /1	Single /1	-	-	Feed-back	-
2022	Engineering	Shortest path problem ⁷⁵ /simulated case + real fuel-pipe routing case	5	100-mm cube simulation + real case	N /N	ACO	Path	Nodewise	Pareto	Shared /1	Static /1	Single /1	-	-	-	-
2022	Transportation	Vehicle routing problem (VRP) ⁷⁶ /25 generated instances + Beijing-Tianjin-Hebei real case	2	25 generated instances + 1 real case	Y /Y	ACS	Routes	Nodewise	Pareto	Shared /1	Static /1	Obj.-wise + Cooperative /2	Local-search /trajectory	-	-	Filter
2022	Informatics	Focused crawling ⁷⁷ /live web pages on typhoon/storm disasters	4	15,000 web-pages/topic	N /N	ACO	Path	Nodewise	Pareto	Shared /1	- /0	Single /1	-	-	Feed-back	Filter
2022	Transportation	Orienteering problem ⁷⁸ /Kyoto/Sapporo/Tokyo trip records + user evaluation	3	100–150 POIs	N /Y	ACO	Path	Assignment	Pareto	Shared /1	Static /1	Single /1	-	-	Feed-back	Filter
2022	Transportation	Orienteering problem ⁷⁹ /Xiamen mobile-signaling data (100,000 users)	2	100,000 users	N /N	-	Path	Nodewise	Scalarization	Shared /1	Static /1	- /-	-	-	-	Filter
2022	Engineering	Multiple traveling salesman problem (mTSP) ⁸⁰ /TSPLIB-derived simulation + hardware-in-the-loop multi-UAV validation	2	50 tasks	Y /Y	ACS	Routes	Assignment	Pareto	Obj.-wise /m	Multi-matrix /V	Cooperative /24	-	-	-	-
2022	Transportation	Vehicle routing problem (VRP) ⁸¹ /benchmark instances + Guangzhou/Shenzhen transport data	3	-	Y /Y	ACS	Routes	Nodewise	Pareto	Obj.-wise /3	Static /1	Obj.-wise + Cooperative /3	Local-search /trajectory	-	-	Filter
2022	Transportation	Shortest path problem ⁸² /ERA-Interim sea-ice data (August 2018)	2	-	N /Y	ACO	Path	Nodewise	Pareto	Shared /1	Static /1	Single /1	-	-	-	Filter
2023	Transportation	Orienteering problem ⁸³ /96 Guangzhou POIs + Baidu Map travel times/fares	2	96 POIs	N /Y	ACS	Composite	Staged	Pareto	Obj.-wise /2	Static /1	Obj.-wise + Cooperative /2	Operators /meta-heuristic	-	-	Filter

Table SM-3: Extended MOACO application classification — continued.

Year	Domain	Canonical problem /Specific application	#O	Size	Eq. /Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2024	Transportation	Vehicle routing problem (VRP) ⁸⁴ /MOGenConVRP-derived instances	3	-	N /Y	ACO	Routes	Assignment	Indicator	Colony-wise /3	- /0	Heterogeneous + Cooperative /3	Operators /meta-heuristic	-	-	-
2024	Engineering	Mobile robot path planning ⁸⁵ /virtual grid environments	3	20×20– 30×30 grids	N /N	ACO	Path	Nodewise	Scalarization	Shared /1	Static /1	Single /1	-	-	Feed-back	-
2025	Transportation	Vehicle routing problem (VRP) ⁴⁶ /Solomon RC1 MOVRPTW	2	-	N /Y	ACO	Routes	Nodewise	Pareto	Shared /1	- /0	Single /1	Local-search /trajectory	-	Feed-back	Hybrid
2025	Transportation	Multiple traveling salesman problem (mTSP) ⁸⁶ /Guangzhou traffic data from Baidu Maps	2	100–400 customers 5/10 employees	Y /N	ACS	Routes	Assignment	Pareto	Shared /1	Static /1	Single /1	Local-search /trajectory	-	Scheduled	-
2025	Communications	Shortest path problem ⁴³ /10×10×10 3-D grid with random obstacles	3	10×10×10 grid	N /N	ACO	Path	Nodewise	Scalarization	Shared /1	Mixed /2	Single /1	-	Policy /DQN	Feed-back	Filter
2025	Transportation	Shortest path problem ⁴² /real Arctic ice-chart imagery; 12 evaluated dates	3	12 dates 10 km × 10 km grid cell	N /Y	ACO	Path	Nodewise	Scalarization	Shared /1	Static /1	Single /1	-	Policy /Q-learning	Feed-back	Filter
2026	Informatics	Resource-constrained path problem ⁸⁷ /BioASQ 8b + HotpotQA; controlled noise/contradictions	5	3,200 / 113k questions	N /Y	ACO	Path	Nodewise	Scalarization	Shared /1	Learned /1	Single /1	-	-	Feed-back	-
2026	Transportation	Traveling salesman problem (TSP) ⁸⁸ /five-day field study of 10 delivery drivers (GPS/physiology/environment)	2	10 drivers	N /N	ACO	Routes	Nodewise	Pareto	Shared /1	Static /1	Single /1	-	-	-	-
2026	Transportation	Multiple traveling salesman problem (mTSP) ⁸⁹ /GTA-20/Urban-35 real-data cases + Synthetic-50/75	4	20/35 50/75 nodes	Y /Y	Mixed	Routes	Nodewise	Scalarization	Obj.-wise /4	Dynamic /adaptive /1	Single /1	Mixed ^b	-	Feed-back	Filter
2026	Transportation	AUV path planning ⁹⁰ /synthetic maps + real nautical charts + field tests with a real AUV	3	40×40 grid 8 field targets	N /Y	ACO	Path	Nodewise	Scalarization	Shared /1	Static /1	Single /1	-	-	Feed-back	-
2026	Communications	Multicast routing problem ⁴⁷ /three multicast benchmark networks (Net 1–3)	4–6	3 benchmark networks	Y /Y	ACO	Tree	Nodewise	Pareto	Mixed /V	Multi-matrix /V	Single /1	-	-	-	Filter
2026	Transportation	Vehicle routing problem (VRP) ⁴⁸ /Solomon C101–C108	3	-	N /Y	ACO	Routes	Nodewise	Pareto	Shared /1	Dynamic /adaptive /1	Single /1	-	-	Feed-back	-
2026	Communications	Wireless sensor network routing ⁹¹ /100-node WSN attack scenarios	2	100 nodes	N /Y	ACO	Path	Nodewise	Pareto	Obj.-wise /2	Static /1	Single /1	-	-	-	Filter
2026	Communications	Shortest path problem ⁹² /SUMO/NS-3 on a real Tangier road map	4	30–150 vehicles	N /N	ACO	Path	Nodewise	Scalarization	- /-	- /0	- /-	-	-	-	-

2 — Scheduling and sequencing

Table SM-3: Extended MOACO application classification — continued.

Year	Domain	Canonical problem /Specific application	#O	Size	Eq. /Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2001	Manufacturing	Single-machine scheduling ⁹³ /aluminum continuous-casting plant	4	-	N /Y	AS	Permutation	Positional	Lexicographic	Shared /1	Static /1	Single /1	-	-	-	Filter
2006	Manufacturing	Assembly line balancing ⁹⁴ /benchmark instance(s); details not reported in source paper	4	21–74 tasks	N /N	Mixed	Assignment	Sampling	Scalarization	Shared /1	Static /1	Single /1	-	-	-	Filter
2007	Manufacturing	Flexible job shop scheduling ¹² /15 jobs, 56 operations, 10 machines	3	15 jobs / 56 ops. / 10 machines	N /N	MMAS	Assignment	Assignment	Scalarization	Comp.-wise /2	Static /1	Single /1	Local-search /trajectory	-	-	Filter
2010	Manufacturing	Assembly line balancing ⁹⁵ /published 25-piece instance extended to three objectives	3	25 pieces	Y /Y	ACS	Assignment	Assignment	Pareto	Shared /1	Static /1	Single /1	-	-	-	Filter
2011	Manufacturing	Assembly line balancing ⁹⁶ /10 realistic instances + Nissan Barcelona plant	2	10 instances + Nissan case	Y /Y	ACS	Assignment	Assignment	Pareto	Shared /1	Static /1	Heterogeneous + Cooperative /5	-	-	Feedback	Filter
2011	Manufacturing	Assembly line balancing ⁹⁷ /24 adapted reference problems with random times and demand scenarios	3	7–148 tasks	N /Y	ACS	Assignment	Assignment	Scalarization	Shared /1	Static /1	Single /1	Local-search /trajectory	-	-	Filter
2013	Manufacturing	Assembly line balancing ⁹⁸ /nine adapted benchmarks + Nissan Barcelona plant	2	10 instances Nissan = 140 tasks	Y /Y	Mixed	Assignment	Assignment	Pareto	Mixed /V	Static /1	Mixed /V	-	Policy /Q-learning	Feedback	Filter
2013	Water	Reservoir operation ⁹⁹ /Murray River, South Australia	2–3	-	Y /Y	Mixed	Sequence	Positional	Scalarization	Mixed /V	Static /1	Single /1	-	-	-	-
2014	Biomedical	Operating room scheduling ¹⁰⁰ /MD Anderson: 28 surgeries, 17 operating rooms	3	28 surgeries / 17 ORs	Y /Y	ACO	Assignment	Assignment	Pareto	Comp.-wise /3	Static /1	Single /1	-	-	-	Filter
2015	Manufacturing	Assembly line balancing ¹⁰¹ /three realistic instances + Nissan Pathfinder industrial case	2	140 tasks / 378 operations	Y /Y	ACS	Assignment	Assignment	Pareto	Shared /1	Static /1	Mixed /V	Local-search /trajectory	-	Feedback	Filter
2015	Computing	Cloud task scheduling ¹⁰² /CloudSim simulation	2	100–600 tasks 1000 VMs	Y /Y	ACO	Assignment	Assignment	Scalarization	Shared /1	Static /1	Single /1	-	-	-	Filter
2017	Transportation	Train rescheduling ¹⁰³ /UK rail junction, timetable, and physical data	2	9 scenarios 2/5/8 trains	N /N	PACO	Permutation	Positional	Pareto	Obj.-wise /m	- /0	Single /1	-	-	-	Hybrid
2017	Manufacturing	Job shop scheduling ¹⁰⁴ /100 synthetic instances + 30 real print-shop records	3	100 synthetic + 30 real cases	Y /Y	ACO	Sequence	Decoding	Scalarization	Comp.-wise /2	Static /1	Single /1	Mixed ^c	-	Feedback	Decoder
2018	Manufacturing	Flexible job shop scheduling ¹⁰⁵ /random 10-part/6-work-center instance	5	10 parts / 6 work centers	Y /Y	ACO	Composite	Decoding	Pareto	Mixed /8	Static /1	Single /1	-	-	Adaptive	Decoder
2019	Computing	Workflow scheduling ¹⁰⁶ /real scientific workflows + Amazon EC2 parameters	2	24–1000 workflow tasks	N /N	ACS	Assignment	Assignment	Pareto	Obj.-wise /2	Static /1	Obj.-wise + Cooperative /2	-	-	-	-

Table SM-3: Extended MOACO application classification — continued.

Year	Domain	Canonical problem /Specific application	#O	Size	Eq. /Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2019	Manufacturing	Parallel batch-machine scheduling ¹⁰⁷ /randomly generated batch-machine instances	2	100-job tests	Y /Y	ACO	Assignment	Assignment	Pareto	Obj.-wise /m	Dynamic /adaptive /1	Single /1	Local-search /trajectory	-	-	Filter
2019	Computing	Cloud task scheduling ¹⁰⁸ /CloudSim task-scheduling scenarios	3	10–50 tasks 5–25 VMs	Y /Y	ACO	Assignment	Assignment	Scalarization	Shared /1	Static /1	Single /1	Operators /meta-heuristic	-	-	Filter
2019	Manufacturing	Single-machine scheduling ¹⁰⁹ /random single-machine instances	2	5/10/15/20 jobs	Y /Y	ACO	Permutation	Assignment	Pareto	Shared /1	Static /1	Single /1	Operators /meta-heuristic	-	-	Filter
2021	Computing	Cloud task scheduling ¹¹⁰ /cloud simulation environment	2	10–200 tasks 1,000 requests total	N /Y	ACO	Assignment	Assignment	Scalarization	- /-	Static /1	- /-	Operators /meta-heuristic	-	-	-
2022	Manufacturing	Hybrid flow shop scheduling ¹¹¹ /600 random hybrid-flow-shop instances	2	600 instances	Y /Y	MOEA/D	Assignment	Decoding	Decomposition	Comp.-wise /2	Static /1	Single /1	Mixed ^d	-	-	Decoder
2025	Manufacturing	Vehicle routing problem (VRP) ⁴⁴ /30 synthetic HRSP-DOA instances + steel-company industrial case	2	120–210 initial orders +15–30/event real: 300 initial orders	Y /Y	ACO	Routes	Nodewise	Indicator	Shared /1	Mixed /V	Single /1	-	Selection /Bandit	Learned	Filter
2025	Biomedical	Task allocation ¹¹² /simulated hospital environment	2	-	Y /Y	ACO	Assignment	Assignment	Scalarization	Shared /1	Static /1	Single /1	-	-	Feed-back	Filter
2026	Manufacturing	Flexible job shop scheduling ⁵¹ /realistic cable-production benchmarks with disruptions	4	10–150 orders 100–2700 ops.	Y /Y	ACO	Assignment	Assignment	Pareto	Shared /1	Mixed /2	Single /1	-	- /GNN	Feed-back	Filter
3 — Selection, packing and portfolio																
2004	Planning	Project portfolio selection ¹¹³ /18 random instances + real R&D portfolio case	K 5 /10 tested	18 synthetic + 1 real portfolio	N /Y	ACS	Subset	Subset-build	Pareto	Obj.-wise /m	Static /1	Single /1	-	-	-	Filter
2009	Informatics	Feature selection ¹¹⁴ /four UCI datasets + one artificial dataset	2	9–56 features 32–1000 samples	N /N	ACO	Subset	Subset-build	Scalarization	Comp.-wise /2	Dynamic /adaptive /2	Heterogeneous + Cooperative /2	-	-	-	Filter
2014	Manufacturing	Sensor placement ¹¹⁵ /methanol-plant steam measurement network	3	-	Y /N	ACO	Subset	Subset-build	Preference	Obj.-wise /m	Multi-matrix /m	Single /1	-	-	Adaptive	Filter
2015	Planning	Project portfolio selection ¹¹⁶ /100-project literature instance	7–10 10 tested	100 projects	N /Y	ACS	Subset	Subset-build	Pareto	Shared /1	Static /1	Single /1	Local-search /trajectory	-	-	Repair
2015	Biomedical	Feature selection ¹¹⁷ /simulated epistasis + real Alzheimer GWAS	2	-	N /N	ACS	Subset	Subset-build	Pareto	Shared /1	- /0	Single /1	-	-	-	Filter
2015	Informatics	Next Release Problem (NRP) ¹¹⁸ /two expert-supplied NRP instances	2	-	Y /Y	ACS	Subset	Subset-build	Decomposition	Obj.-wise /2	Static /1	Single /1	-	-	-	Filter

Table SM-3: Extended MOACO application classification — continued.

Year	Domain	Canonical problem /Specific application	#O	Size	Eq. /Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2018	Planning	Project portfolio selection ¹¹⁹ /illustrative tabular example; no experimental validation	3	-	N /N	ACO	Subset	Assign- ment	Preference	Obj.-wise /3	Multi- matrix /m	Obj.-wise + Coopera- tive /3	-	-	-	Decoder
2019	Informatics	Cardinality-constrained subset selection ¹²⁰ /4,856 Twitter comments (#pilpres)	2	4,856 comments	Y /Y	AS	Subset	Subset- build	Pareto	Obj.-wise /m	Static /1	Single /1	-	-	-	Filter
2021	Computing	Knapsack problem ¹²¹ /inherited Wang et al. knapsack simulation instances	3	20 items	N /Y	ACO	Subset	Subset- build	Scalariza- tion	Shared /1	Static /1	Single /1	Operators /meta- heuris- tic	-	-	-
2023	Informatics	Feature selection ³⁹ /13 classification datasets including real sets + Madelon	2	18–5147 features 72–2600 samples	N /N	Binary- ACO	Subset	Subset- build	Pareto	Shared /1	Static /1	Single /1	-	-	-	Filter
2026	Informatics	Extractive text summarization ¹²² /TyHagupit, SandyHShoot, HydBlast, UFlood disaster tweets	4	4 tweet datasets	N /N	ACO _R	Subset	Subset- build	Lexico- graphic	Shared /1	Learned /1	Single /1	-	-	-	-
2026	Communica- tions	Cluster head selection ⁵⁰ /MATLAB WSN deployments with 50–200 random nodes	3	50–200 nodes	N /N	ACO	Subset	Assign- ment	Scalariza- tion	Shared /1	Static /1	Single /1	-	Policy /DQN	-	-
4 — Assignment and resource allocation																
2007	Engineering	Multidimensional assignment problem ¹²³ /3-/5-target scenarios with noise and clutter	2	3/5 targets	N /N	ACS	Assign- ment	Assign- ment	Pareto	Shared /1	Static /1	Hetero- geneous + Coopera- tive /m + 1	-	-	-	-
2010	Computing	Heterogeneous parallel- machine scheduling ¹²⁴ /Shanghai education/research network topology	3	60–300 jobs 16 hosts	Y /Y	ACS	Assign- ment	Assign- ment	Pareto	Shared /1	Static /1	Single /1	Operators /meta- heuris- tic	-	-	Filter
2013	Power	Voltage/reactive-power control ¹²⁵ /33- and 69-bus test systems	5	33/69 buses	Y /Y	Mixed	Assign- ment	Decoding	Lexico- graphic	Shared /1	Static /1	Single /1	-	-	-	Decoder
2017	Other	Network flow problem ¹²⁶ /FAOSTAT trade data + simulated worst-case regional shock	2	74 trading partners	Y /Y	PACO	Assign- ment	Assign- ment	Pareto	Obj.-wise /m	- /0	Single /1	-	-	-	-
2017	Communica- tions	Radio resource allocation ¹²⁷ /LTE-Sim scenarios	3	5–25 femtocells	N /N	Binary- ACO	Assign- ment	Assign- ment	Scalariza- tion	Shared /1	- /0	Single /1	-	-	-	-
2017	Computing	Virtual machine placement ¹²⁸ /real workloads/data evaluated in CloudSim	2	100–500 VMs	Y /Y	ACS	Assign- ment	Assign- ment	Pareto	Shared /1	Static /1	Single /1	-	-	-	Filter
2017	Power	Economic emission dispatch / optimal power flow ¹²⁹ /standard synthetic power-system test cases	2	6/10 units IEEE 30-bus	Y /Y	ACO _R	Assign- ment	Sampling	Pareto	Shared /1	- /0	Hetero- geneous + Coopera- tive /2	Mixed ^a	-	-	Hybrid
2018	Computing	Virtual machine placement ¹³⁰ /random instances on a simulated Fat-Tree network	2	300 VMs / 300 PMs 3000 instances	Y /Y	ACS	Assign- ment	Assign- ment	Preference	Shared /1	Static /1	Single /1	-	-	-	Filter

Table SM-3: Extended MOACO application classification — continued.

Year	Domain	Canonical problem /Specific application	#O	Size	Eq. /Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2018	Computing	Heterogeneous parallel-machine scheduling ¹³¹ /eight CloudSim instances	2	5–25 tasks + 150-task case	N /N	ACO	Assign- ment	Assign- ment	Scalariza- tion	Shared /1	Static /1	Single /1	-	-	-	-
2019	Computing	Virtual machine consolidation ¹³² /PlanetLab workloads in CloudSim	3	800 hosts 41000 VMs	N /Y	ACS	Assign- ment	Assign- ment	Scalariza- tion	Shared /1	Static /1	Single /1	-	-	-	Filter
2020	Transporta- tion	Task allocation ¹³³ /Beijing T-Drive taxi trajectories	2	20–40 tasks >10,000 taxis	N /Y	ACO	Assign- ment	Assign- ment	Pareto	Shared /1	Static /1	Single /1	-	-	-	Filter
2020	Computing	Resource allocation ³³ /simulated resource-allocation scenarios; no real dataset identified	-	550 nodes	N /N	ACO	Assign- ment	Assign- ment	Pareto	- /-	Static /1	- /-	Local- search /trajec- tory	Policy /RL	-	-
2021	Transporta- tion	Crew rostering problem ¹³⁴ /seven monthly crew-roster instances from a North American airline	2	7 monthly rosters	Y /Y	ACS	Assign- ment	Assign- ment	Pareto	Obj.-wise /2	Mixed /3	Obj.-wise + Coopera- tive /2	Local- search /trajec- tory	-	-	Filter
2023	Computing	Heterogeneous parallel-machine scheduling ¹³⁵ /CloudSim heterogeneous-machine scenarios	4	100–2000 tasks 10–50 VMs	N /N	ACO	Assign- ment	Assign- ment	Pareto	Shared /1	Multi- matrix /4	Single /1	-	-	Feed- back	Filter
5 — Design and configuration																
2009	Engineering	Composite laminate structural design ¹³⁶ /numerical laminate-design instances from literature models	2	8/16/28 layers	Y /Y	ACS	Sequence	Positional	Scalariza- tion	Shared /1	- /0	Single /1	-	-	-	Filter
2009	Communica- tions	Antenna topology optimization ¹³⁷ /NEC electromagnetic model	2	50 grid points	N /N	ACS	Path	Nodewise	Pareto	Shared /1	- /0	Single /1	Local- search /trajec- tory	-	-	Filter
2011	Manufactur- ing	Supply chain network design ¹³⁸ /bulldozer supply-chain instance and variants	2	38 activities 105 resource options	Y /N	ACO	Assign- ment	Assign- ment	Pareto	Mixed /V	Static /1	Coopera- tive /30	-	-	-	Filter
2011	Engineering	Linear motor design ¹³⁹ /analytical electromagnetic models for linear motor and solenoid	P1: 2 P2: 2	-	N /Y	ACO	Continu- ous	Levelwise	Pareto	Obj.-wise /m	Static /1	Obj.-wise + Coopera- tive /m	Mixed ^f	-	-	-
2015	Engineering	Truss sizing optimization ¹⁴⁰ /10/25/60/72/942-bar truss instances	2	10/25/60 72/942 bars	N /Y	ACS	Assign- ment	Assign- ment	Pareto	Obj.-wise /m	- /0	Single /1	-	-	Feed- back	-
2015	Biomedical	RNA inverse folding ¹⁴¹ /PseudoBase++ biological structures	3	249 test structures 7,470 designs	N /N	ACO	Sequence	Nodewise	Scalariza- tion	Shared /1	- /0	Single /1	-	-	-	-
2015	Computing	Web service composition ¹⁴² /randomly generated service-composition scenarios	2	15 scenarios	Y /Y	ACS	Path	Nodewise	Pareto	- /-	Static /1	Single /1	-	-	-	-
2015	Engineering	Topology optimization ¹⁴³ /realistic C-core magnetic actuator with finite-element model	2	-	Y /Y	ACO	Configu- ration	Nodewise	Pareto	Shared /1	Static /1	Single /1	Mixed ^g	-	-	-

Table SM-3: Extended MOACO application classification — continued.

Year	Domain	Canonical problem /Specific application	#O	Size	Eq. /Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2016	Engineering	Structural topology and sizing optimization ¹⁴⁴ /10- and 40-story braced frames	2	10/40 stories	N /Y	ACO	Discrete	Positional	Scalariza- tion	Shared /1	- /0	Single /1	Operators /meta- heuristic	-	-	-
2019	Communica- tions	Tree network topology design ¹⁴⁵ /literature network-topology test cases	5	-	N /Y	PACO	Tree	Nodewise	Scalariza- tion	Shared /1	Static /1	Single /1	-	-	-	Filter
2019	Engineering	Neural architecture / hyperparameter optimization ¹⁴⁶ /gasoline-engine test-bench data	3	-	N /Y	ACS	Configu- ration	Positional	Pareto	Shared /1	Dynamic /adaptive /1	Single /1	-	-	-	-
2020	Other	Molecular design ¹⁴⁷ /commercial reagents + real zeolite structures; computational evaluation	2	-	N /Y	MMAS	Configu- ration	Assign- ment	Pareto	Shared /1	Static /1	Single /1	-	-	-	Filter
2022	Biomedical	Primer design ³⁷ /three DNA-template primer-design cases; empirical origin not reported	2	3 primer- design cases	Y /Y	MOEA/D	Permuta- tion	Nodewise	Decompo- sition	Subprob.- wise /K	Static /1	Subprob.- wise + Coopera- tive /K	-	Selection /Learning automata	Learned	Filter
2023	Communica- tions	Network topology design ¹⁴⁸ /simulation + Tianbot Mini robot platform	2	-	N /Y	ACO	Subset	Subset- build	Pareto	Obj.-wise /2	Static /1	Hetero- geneous + Coopera- tive /3	-	-	-	-
2026	Informatics	Web service composition ¹⁴⁹ /extended EFACO service-composition dataset	4	-	Y /N	PACO	Assign- ment	Assign- ment	Preference	Obj.-wise /4	Static /1	- /-	Operators /meta- heuristic	-	Feed- back	Filter
6 — Continuous and inverse parameter optimization																
2004	Manufactur- ing	Grinding process parameter optimization ¹⁵⁰ /literature-derived numerical case Oblique-shock inverse design	3	4 design variables	N /Y	ACO	Continu- ous	Sampling	Scalariza- tion	Shared /1	- /0	Single /1	Mixed ^h	-	Sched- uled	-
2011	Engineering	Supersonic nozzle shape optimization ¹⁵¹ Airfoil shape optimization /Rankine–Hugoniot, quasi-1D MacCormack, and 2-D CFD/NACA models	P1: 3 P2: 2 P3: 2	-	Y /Y	ACO	Continu- ous	Levelwise	Preference	Variable- wise /1	Dynamic /adaptive /1	Single /1	-	-	Feed- back	-
2018	Engineering	Gas pipeline network optimization ²⁸ /ZDT1–ZDT3 + modeled gas-company pipeline case	2	-	Y /Y	ACO	Continu- ous	Sampling	Scalariza- tion	Shared /1	- /0	Single /1	-	-	-	-
2018	Water	Water-distribution-system leak detection / inverse calibration ¹⁵² /two benchmark water networks with simulated leaks	2	2 benchmark networks	N /Y	ACO	Continu- ous	Sampling	Scalariza- tion	Shared /1	Dynamic /adaptive /1	Single /1	-	-	-	-
2019	Power	Passive harmonic filter design ¹⁵³ /randomly selected harmonic-source scenarios	4	-	N /Y	ACO	Continu- ous	Sampling	Pareto	Variable- wise /1	- /0	- /-	-	-	Feed- back	-

Table SM-3: Extended MOACO application classification — continued.

Year	Domain	Canonical problem /Specific application	#O	Size	Eq. /Ineq.	SE	SR	SC	MO	PM /#P	HI /#H	CA /#C	HY	LMR /LMM	PA	FP
2024	Water	Reservoir operation ⁴⁰ /Three Gorges Reservoir model calibrated with observed flow/temperature	3	-	Y /Y	ACO _R	Continu- ous	Sampling	Pareto	Variable- wise /1	- /0	Single /1	Mixed ¹	-	Feed- back	-
7 — Location and layout																
2005	Other	Facility location ¹⁵⁴ /Singapore fire-station siting (GIS/SCDF data)	3	200-m / 25-m GIS grids	Y /N	Mixed	Subset	Subset- build	Pareto	Shared /1	Static /1	Single /1	Local- search /trajec- tory	-	-	Filter
2009	Power	Network device placement ¹⁵⁵ /22-kV Thailand PEA feeder	3	51 sections / 44 load points	N /Y	ACS	Sequence	Positional	Pareto	Shared /1	Multi- matrix /3	Single /1	-	-	Feed- back	Filter
2019	Manufactur- ing	Facility layout ¹⁵⁶ /literature facility-layout benchmarks	2	-	Y /Y	ACO	Continu- ous	Sampling	Pareto	Shared /1	Static /1	Single /1	Mixed ¹	-	Feed- back	Repair
2026	Communica- tions	Maximum coverage location problem ¹⁵⁷ /161 real airport/VDL-M2 sites + 1,206 ATS points	2	161 sites / 1,206 points	Y /Y	ACO	Binary	Bitwise	Pareto	Shared /1	Dynamic /adaptive /1	Single /1	Local- search /trajec- tory	-	Feed- back	Filter
8 — Compound problems																
2014	Planning	Resource-constrained project scheduling (RCPSP) + resource assignment ¹⁵⁸ /literature examples + 27 generated RCPSP instances	2	27 generated instances	Y /Y	ACS	Assign- ment	Assign- ment	Pareto	Shared /1	Static /1	Obj.-wise + Coopera- tive /2	Local- search /trajec- tory	-	-	Filter
2019	Transporta- tion	Facility location + vehicle routing problem ¹⁵⁹ /23 Guangdong cities + Google Maps distances	2	23 cities	Y /Y	ACO	Routes	Nodewise	Pareto	Obj.-wise /m	Static /1	Single /1	Operators /meta- heuris- tic	-	-	Filter
2021	Transporta- tion	Traveling salesman problem (TSP) + shortest path problem ¹⁶⁰ /Fahadan (Yazd) pedestrian network with real urban attributes	4	-	N /N	ACO	Path	Nodewise	Pareto	Shared /1	Static /1	Single /1	-	-	-	-
9 — Graph partitioning and clustering																
2019	Informatics	Community detection ¹⁶¹ /LFR synthetic networks + eight real networks	2	100–1000 nodes (LFR) + 8 real networks	N /N	MOEA/D	Locus	Decoding	Decompo- sition	Subprob.- wise /K	Static /1	Single /1	Mixed ^k	-	Feed- back	Decoder
2020	Informatics	Community detection ³⁴ /synthetic networks + four real networks	2	34/62 105/115 nodes (real) 128-node GN	N /N	MOEA/D	Locus	Decoding	Decompo- sition	Subprob.- wise /K	Static /1	Subprob.- wise + Coopera- tive /K	Mixed ¹	-	Feed- back	Decoder
2020	Informatics	Community detection ¹⁶² /extended GN benchmark + seven real networks	2	+ 7 real networks	N /N	ACS	Locus	Decoding	Pareto	Shared /1	- /0	Single /1	-	-	-	Decoder

Mixed-category notes (HY). ^a 3-opt tour search + k-opt subpath search. ^b Local route improvement + adaptive multi-criteria control. ^c Local search + mutation + PSO-based adaptive parameter tuning. ^d ACO within the MOEA/D framework + problem-specific neighborhood improvement. ^e Elite/normal search populations + niche-based diversity search. ^f Steady-state genetic operators + dynamic pattern-search local search. ^g MOACO topology search + NSGA-II shape refinement. ^h Local/global continuous search + mutation/diffusion. ⁱ ϵ -MOEA design elements + ACO _{$\mathbb{R}$} local search. ^j Adaptive-gradient local search + heuristic department-deformation/niching operations. ^k Strong-community improvement + neighbor-based solution improvement across decomposition groups. ^l MOEA/D framework + weighted simulated-annealing local search.

References

1. L. Gambardella, É. Taillard, and G. Agazzi. Macs-vrptw: A multiple ant colony system for vehicle routing problems with time windows. Technical report, Istituto Dalle Molle Di Studi Sull Intelligenza Artificiale, (1999).
2. C. E. Mariano and E. M. Morales. MOAQ an Ant-Q algorithm for multiple objective optimization problems. In *Proceedings of the 1st Annual Conference on Genetic and Evolutionary Computation - Volume 1*, GECCO'99, pp. 894–901, San Francisco, CA, USA, (1999). Morgan Kaufmann Publishers Inc. ISBN 1558606114.
3. S. Iredi, D. Merkle, and M. Middendorf. Bi-criterion optimization with multi colony ant algorithms. In eds. E. Zitzler, L. Thiele, K. Deb, C. A. Coello Coello, and D. Corne, *Evolutionary Multi-Criterion Optimization*, pp. 359–372, Berlin, Heidelberg, (2001). Springer Berlin Heidelberg. ISBN 978-3-540-44719-1. doi: 10.1007/3-540-44719-9_25.
4. V. T'kindt, N. Monmarché, F. Tercinet, and D. Laügt, An ant colony optimization algorithm to solve a 2-machine bicriteria flowshop scheduling problem, *European Journal of Operational Research*. **142**(2), 250–257, (2002). ISSN 0377-2217. doi: [https://doi.org/10.1016/S0377-2217\(02\)00265-5](https://doi.org/10.1016/S0377-2217(02)00265-5).
5. P. Cardoso, M. Jesus, and A. Márquez. MONACO: Multi-objective network optimisation based on an ACO. In *Proceedings of the X Encuentros de Geometría Computacional*, Seville, Spain, (2003).
6. M. Guntsch and M. Middendorf. Solving multi-criteria optimization problems with population-based aco. In eds. C. M. Fonseca, P. J. Fleming, E. Zitzler, L. Thiele, and K. Deb, *Evolutionary Multi-Criterion Optimization*, pp. 464–478, Berlin, Heidelberg, (2003). Springer Berlin Heidelberg. ISBN 978-3-540-36970-7. doi: 10.1007/3-540-36970-8_33.
7. M. López-Ibáñez, L. Paquete, and T. Stützle. On the design of aco for the biobjective quadratic assignment problem. In eds. M. Dorigo, M. Birattari, C. Blum, L. M. Gambardella, F. Mondada, and T. Stützle, *Ant Colony Optimization and Swarm Intelligence*, pp. 214–225, Berlin, Heidelberg, (2004). Springer Berlin Heidelberg. ISBN 978-3-540-28646-2. doi: 10.1007/978-3-540-28646-2_19.
8. I. Alaya, C. Solnon, and K. Ghédira. Ant Colony Optimization for Multi-Objective Optimization Problems. In *19th IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2007)*, vol. 1, pp. 450–457, (2007). doi: 10.1109/ICTAI.2007.108.
9. D. Angus. Crowding population-based ant colony optimisation for the multi-objective travelling salesman problem. In *IEEE Symposium on Computational Intelligence in Multicriteria Decision Making*, (2007). doi: 10.1109/MCDM.2007.369110.
10. D. Angus. Population-based ant colony optimisation for multi-objective function optimisation. In *Progress in Artificial Life*, pp. 232–244. Springer, (2007). doi: 10.1007/978-3-540-76931-6_22.
11. A. García-Nájera and J. A. Bullinaria. Extending acor to solve multi-objective problems. In *Proceedings of the UK Workshop on Computational Intelligence*, (2007).
12. L.-N. Xing, Y.-W. Chen, and K.-W. Yang. Interactive Fuzzy Multi-objective Ant Colony Optimization with Linguistically Quantified Decision Functions for Flexible Job Shop Scheduling Problems. In *2007 Frontiers in the Convergence of Bioscience and Information Technologies (FBIT 2007)*, pp. 801–806, (2007). doi: 10.1109/FBIT.2007.18.
13. B. Yagmahan and M. M. Yenisey, Ant colony optimization for multi-objective flow shop scheduling problem, *Computers & Industrial Engineering*. **54**, 411–420, (2008). doi: 10.1016/j.cie.2007.08.003.
14. L. T. Bui, J. M. Whitacre, and H. A. Abbass. Performance analysis of elitism in multi-objective ant colony optimization algorithms. In *2008 IEEE Congress on Evolutionary Computation (IEEE World Congress on Computational Intelligence)*, pp. 1633–1640, (2008). doi: 10.1109/CEC.2008.4631010.
15. S. Häckel, M. Fischer, D. Zechel, and T. Teich. A multi-objective ant colony approach for pareto-optimization using dynamic programming. In *Proceedings of the Genetic and Evolutionary Computation Conference*, pp. 33–40, (2008). doi: 10.1145/1389095.1389102.
16. B. Y. M. M. Yenisey, A multi-objective ant colony system algorithm for flow shop scheduling problem, *Expert Systems with Applications*. **37**(2), 1361–1368, (2010). doi: 10.1016/j.eswa.2009.06.105.
17. H. Panahi and R. Tavakkoli-Moghaddam, Solving a multi-objective open shop scheduling problem by a novel hybrid ant colony optimization, *Expert Systems with Applications*. **38**(3), 2817–2822, (2011). doi: 10.1016/j.eswa.2010.08.073.
18. J. Cheng, G. Zhang, Z. Li, and Y. Li, Multi-objective ant colony optimization based on decomposition for bi-objective traveling salesman problems, *Soft Computing*. **16**(4), 597–614, (2012). doi: 10.1007/s00500-011-0759-3.
19. M. López-Ibáñez and T. Stützle, An experimental analysis of design choices of multi-objective ant colony optimization algorithms, *Swarm Intelligence*. **6**(3), 207–232, (2012). doi: 10.1007/s11721-012-0070-7.
20. L. Ke, Q. Zhang, and R. Battiti, Moea/d-aco: A multiobjective evolutionary algorithm using decomposition and ant colony, *IEEE Transactions on Cybernetics*. **43**(6),

- 1845–1859, (2013). doi: 10.1109/TSMCB.2012.2231860.
21. A. M. Mora, P. García-Sánchez, J. J. Merelo, and P. A. Castillo, Pareto-based multi-colony multi-objective ant colony optimization algorithms: An island model proposal, *Soft Computing*. **17**, 1175–1207, (2013). doi: 10.1007/s00500-013-0993-y.
 22. C. Özkale and A. Fıǧlalı, Evaluation of the multiobjective ant colony algorithm performances on biobjective quadratic assignment problems, *Applied Mathematical Modelling*. **37**, 7822–7838, (2013). doi: 10.1016/j.apm.2013.01.045.
 23. I. D. I. D. Ariyasingha and T. G. I. Fernando, Performance analysis of the multi-objective ant colony optimization algorithms for the traveling salesman problem, *Swarm and Evolutionary Computation*. **23**, 11–26, (2015). doi: 10.1016/j.swevo.2015.02.003.
 24. I. B. Mansour and I. Alaya, Indicator based ant colony optimization for multi-objective knapsack problem, *Procedia - Procedia Computer Science*. **60**, 448–457, (2015). doi: 10.1016/j.procs.2015.08.165.
 25. Z. Zhang, C. Gao, Y. Lu, Y. Liu, and M. Liang, Multi-objective ant colony optimization based on the physarum-inspired mathematical model for bi-objective traveling salesman problems, *PLoS ONE*. **11**(1), e0146709, (2016). doi: 10.1371/journal.pone.0146709.
 26. J. Falcón-Cardona and C. Coello Coello, A new indicator-based many-objective ant colony optimizer for continuous search spaces., *Swarm Intelligence*. (2017). doi: 10.1007/s11721-017-0133-x.
 27. Y. guang Zhong and B. Ai, A modified ant colony optimization algorithm for multi-objective assembly line balancing, *Soft Comput.* **21**, 6881–6894, (2017). doi: 10.1007/s00500-016-2240-9.
 28. A. K. Arya and S. Honwad, Multiobjective optimization of a gas pipeline network: an ant colony approach, *Journal of Petroleum Exploration and Production Technology*. **8**, 1389–1400, (2018). doi: 10.1007/s13202-017-0410-7.
 29. Y. Liu, H. Zhou, Y. Wang, X. Ren, and X. Diao, Ant colony optimisation algorithm for multiobjective subset selection problems, *Electronics Letters*. **55**, 1283–1286, (2019). doi: 10.1049/el.2019.1933.
 30. M. L. Fei Li and G. Xu, A quantum ant colony multi-objective routing algorithm in wsn and its application in a manufacturing environment, *Sensors*. **19**, 3334, (2019). doi: 10.3390/s19153334.
 31. I. B. Mansour, I. Alaya, and M. Tagina, A gradual weight-based ant colony approach for solving the multiobjective multidimensional knapsack problem, *Evolutionary Intelligence*. **12**, 253–272, (2019). doi: 10.1007/s12065-019-00222-9.
 32. P. S. Jiaxu Ning, Changsheng Zhang and Y. Feng, Comparative study of ant colony algorithms for multi-objective optimization, *Information*. **10**(1), 11, (2019). doi: 10.3390/info10010011.
 33. S. Vimal, M. Khari, N. Dey, R. G. Crespo, and Y. H. Robinson, Enhanced resource allocation in mobile edge computing using reinforcement learning based moaco algorithm for iiot, *Computer Communications*. **151**, 355–364, (2020). doi: 10.1016/j.comcom.2020.01.018.
 34. P. Ji, S. Zhang, and Z. Zhou, A decomposition-based ant colony optimization algorithm for the multi-objective community detection, *Journal of Ambient Intelligence and Humanized Computing*. **11**, 173–188, (2020). doi: 10.1007/s12652-019-01241-1.
 35. J. Bagherinejad, F. Jolai, R. Abdollahnejad, and M. Shoeib, A hybrid algorithm based on non-dominated sorting ant colony and genetic algorithms for solving multi-objective multi-mode project scheduling problems under resource constraints, *Management and Production Engineering Review*. **11**(2), 88–98, (2020). doi: 10.24425/ mper.2020.133731.
 36. H. Zhao, C. Zhang, and B. Zhang, A decomposition-based many-objective ant colony optimization algorithm with adaptive reference points, *Information Sciences*. **540**, 435–448, (2020). doi: 10.1016/j.ins.2020.06.028.
 37. H. Zhao and C. Zhang, An ant colony optimization algorithm with evolutionary experience-guided pheromone updating strategies for multi-objective optimization, *Expert Systems with Applications*. **201**, 117151, (2022). doi: 10.1016/j.eswa.2022.117151.
 38. H. Zhao, C. Zhang, X. Zheng, C. Zhang, and B. Zhang, A decomposition-based many-objective ant colony optimization algorithm with adaptive solution construction and selection approaches, *Swarm and Evolutionary Computation*. **68**, 100977, (2022). doi: 10.1016/j.swevo.2021.100977.
 39. Z. Wang, S. Gao, M. Zhou, S. Sato, J. Cheng, and J. Wang, Information-Theory-based Nondominated Sorting Ant Colony Optimization for Multiobjective Feature Selection in Classification, *IEEE Transactions on Cybernetics*. **53**(8), 5276–5289, (2023). doi: 10.1109/TCYB.2022.3185554.
 40. R. Qiu, D. Wang, V. P. Singh, Y. Wang, and J. Wu, Integration of deep learning and improved multi-objective algorithm to optimize reservoir operation for balancing human and downstream ecological needs, *Water Research*. **253**, 121314, (2024). doi: 10.1016/j.watres.2024.121314.
 41. R. H. Tamayo, J. Falcón-Cardona, and C. Coello Coello. Reaching pareto front shape invariance with a continuous multi-objective ant colony optimization algorithm.

- In eds. M. Affenzeller, S. M. Winkler, A. V. Kononova, H. Trautmann, T. Tušar, P. Machado, and T. Bäck, *Parallel Problem Solving from Nature – PPSN XVIII*, pp. 252–267, Cham, (2024). Springer Nature Switzerland. ISBN 978-3-031-70085-9.
42. J. Zhou, X. Liu, H. Xie, and Y. Yin, Integrated improved ant colony optimization and Q-learning for ice navigation route planning, *Ocean Engineering*. **342**, 122935, (2025). ISSN 0029-8018. doi: <https://doi.org/10.1016/j.oceaneng.2025.122935>.
 43. X. Zhang, X. Yu, and X. Chen, D-ACO-Based Path Planning for AUV in UWSN: A Hybrid Approach Combining DQN and ACO, *IEEE Sensors Journal*. **25**(24), 44555–44565, (2025). doi: 10.1109/JSEN.2025.3622250.
 44. S.-L. Jiang, Q. Liu, L.-L. Cao, and L. Sun, An Online Learning-Based mACO Approach for Hot Rolling Scheduling Problems Involving Dynamic Order Arrivals, *IEEE Transactions on Automation Science and Engineering*. **22**, 12322–12335, (2025). doi: 10.1109/TASE.2025.3540042.
 45. I. Alaya, I. BenMansour, and M. Nciri. A Graph Neural Network Enhancing Ant Colony Optimization for Multi-Objective Knapsack Problem. In *2025 IEEE 37th International Conference on Tools with Artificial Intelligence (ICTAI)*, pp. 371–375, (2025). doi: 10.1109/ICTAI66417.2025.00055.
 46. Y. Hou, X. Qin, H. Han, and J. Wang, Multiobjective Ant Colony Optimization Algorithm Based on Dynamic Constraint Evaluation Strategy for Highly Constrained Optimization, *IEEE Transactions on Cybernetics*. **55**(10), 4570–4582, (2025). doi: 10.1109/TCYB.2025.3591275.
 47. T. P. França, L. G. A. Martins, and G. M. B. Oliveira. MACO/NDS: Many-objective Ant Colony Optimization based on Non-Dominated Sets. In *2018 IEEE Congress on Evolutionary Computation (CEC)*, pp. 1–8, (2018). doi: 10.1109/CEC.2018.8477958.
 48. Y. Hou, J. Wang, H. Han, and J. Wang, Multimodal multi-objective ant colony optimization based on ecological niche comprehensive evaluation indicators, *Applied Soft Computing Journal*. **201**, 115468, (2026). doi: 10.1016/j.asoc.2026.115468.
 49. A. Zakraoui and I. B. Mansour. A Generative-AI-Assisted Frequency–Rarity Pheromone Update for Ant Colony Optimization Applied to the MOMKP. In *2026 Intelligence in Business and Industry (IBI)*, pp. 1–7, (2026). doi: 10.1109/IBI68858.2026.11604104.
 50. T. M. and J. A. Baskaradas, A hybrid mo-aco and deep reinforcement learning framework for energy-aware routing in wireless sensor networks, *SSRG International Journal of Electronics and Communication Engineering*. **13**(3), 175–184, (2026). doi: 10.14445/23488549/IJECE-V13I3P114.
 51. S. Tariq, A. Raza, B. A. Chuddher, Preetam, R. U. Hassan, and R. M. Ishaque. A Graph Neural Network Enhanced Multi-Objective Ant Colony Optimization for Dynamic Predictive-Reactive Scheduling in Cable Production. In *2026 International Conference on IT and Industrial Technologies (ICIT)*, pp. 1–6, (2026). doi: 10.1109/ICIT68548.2026.11577731.
 52. B. Barán and M. Schaerer. A multiobjective ant colony system for vehicle routing problem with time windows. In *Proceedings of the 21st IASTED International Multi-Conference on Applied Informatics*, pp. 97–102, Innsbruck, Austria, (2003).
 53. D. Pinto and B. Barán. Solving multiobjective multicast routing problem with a new ant colony optimization approach. In *3rd International Latin American Networking Conference*, pp. 11–19, (2005). doi: 10.1145/1168117.1168120.
 54. A. M. Mora, J. J. M. Guervós, C. Millán, J. Torrecillas, J. L. J. Laredo, and P. Á. C. Valdivieso. Comparing aco algorithms for solving the bi-criteria military path-finding problem. In *Advances in Artificial Life*, pp. 665–674. Springer, (2007). doi: 10.1007/978-3-540-74913-4_68.
 55. J. A. Sabino, J. E. Leal, T. Stützle, and M. Birattari, A multi-objective ant colony optimization method applied to switch engine scheduling in railroad yards, *Pesquisa Operacional*. **30**(2), 487–514, (2010). doi: 10.1590/S0101-74382010000200013.
 56. P. Deepalakshmi and S. Radhakrishnan, An ant colony-based multi objective quality of service routing for mobile ad hoc networks, *EURASIP Journal on Wireless Communications and Networking*. **2011**, 153, (2011). doi: 10.1186/1687-1499-2011-153.
 57. Z. Fang, X. Zong, Q. Li, Q. Li, and S. Xiong, Hierarchical multi-objective evacuation routing in stadium using ant colony optimization approach, *Journal of Transport Geography*. **19**(3), 443–451, (2011). doi: 10.1016/j.jtrangeo.2010.10.001.
 58. X. Wang, Y. Zhao, D. Wang, H. Zhu, and Q. Zhang, Improved multi-objective ant colony optimization algorithm and its application, *Chinese Journal of Mechanical Engineering*. **26**(5), 1031–1040, (2013). doi: 10.3901/CJME.2013.05.1031.
 59. J. J. S. Chávez, J. W. Escobar, and M. G. Echeverri, A multi-objective pareto ant colony algorithm for the multi-depot vehicle routing problem with backhauls, *International Journal of Industrial Engineering Computations*. **7**, 35–48, (2016). doi: 10.5267/j.ijec.2015.8.003.
 60. H. Faroqi and M. S. Mesgari, Performance comparison between the multi-colony and multi-pheromone aco algorithms for solving the multi-objective routing problem in a public transportation network, *The Journal of Navigation*. **69**, 197–210, (2016). doi: 10.1017/S0373463315000594.
 61. X. Chen, P. Zhang, G. Du, and F. Li, Ant colony optimization based memetic algorithm to solve bi-objective multiple traveling salesmen problem for multi-robot systems, *IEEE Access*. **6**, 21745–21757, (2018). doi: 10.1109/ACCESS.2018.2828499.

62. N. afarin Mokhtari and V. Ghezavati, Integration of efficient multi-objective ant-colony and a heuristic method to solve a novel multi-objective mixed load school bus routing model, *Applied Soft Computing*. **68**, 92–109, (2018). doi: 10.1016/j.asoc.2018.03.049.
63. Y. Li, H. Soleimani, and M. Zohal, An improved ant colony optimization algorithm for the multi-depot green vehicle routing problem with multiple objectives, *Journal of Cleaner Production*. **227**, 1161–1172, (2019). doi: 10.1016/j.jclepro.2019.03.185.
64. S. B. Sasi and R. Santhosh, Multiobjective routing protocol for wireless sensor network optimization using ant colony conveyance algorithm, *International Journal of Communication Systems*. **34**(6), 1–13, (2021). doi: 10.1002/dac.4270.
65. Z. Sun, M. Wei, Z. Zhang, and G. Qu, Secure routing protocol based on multi-objective ant-colony-optimization for wireless sensor networks, *Applied Soft Computing Journal*. **77**, 366–375, (2019). doi: 10.1016/j.asoc.2019.01.034.
66. H. Zhang, Q. Zhang, L. Ma, Z. Zhang, and Y. Liu, A hybrid ant colony optimization algorithm for a multi-objective vehicle routing problem with flexible time windows, *Information Sciences*. **490**, 166–190, (2019). doi: 10.1016/j.ins.2019.03.070.
67. A. Y. Hamed, M. H. Alkinani, and M. R. Hassan, Ant colony optimization for multi-objective multicast routing, *Computers, Materials & Continua*. **63**(3), 1159–1173, (2020). doi: 10.32604/cmc.2020.09176.
68. M. L. Mutar, M. A. Burhanuddin, A. S. Hameed, N. Yusof, M. F. Alrifai, and A. A. Mohammed, Multi-Objectives Ant Colony System for Solving Multi-Objectives Capacitated Vehicle Routing Problem, *Journal of Theoretical and Applied Information Technology*. **98**(24), 4014–4027, (2020). ISSN 1992-8645.
69. S. Vimal, M. Khari, R. G. Crespo, L. Kalaivani, N. Dey, and M. Kaliappan, Energy enhancement using multiobjective ant colony optimization with double q learning algorithm for iot based cognitive radio networks, *Computer Communications*. **154**, 481–490, (2020). doi: 10.1016/j.comcom.2020.03.004.
70. B. Zhao, H. Gui, H. Li, and J. Xue, Cold chain logistics path optimization via improved multi-objective ant colony algorithm, *IEEE Access*. **8**, 142977–142995, (2020). doi: 10.1109/ACCESS.2020.3013951.
71. R. K. Goel and R. Maini, Improved multi-ant-colony algorithm for solving multi-objective vehicle routing problems, *Scientia Iranica*. **28**(6), 3412–3428, (2021). doi: 10.24200/sci.2019.51899.2414.
72. G. Zhang, H. Wang, W. Zhao, Z. Guan, and P. Li, Application of improved multi-objective ant colony optimization algorithm in ship weather routing, *Journal of Ocean University of China*. **20**(1), 45–55, (2021). doi: 10.1007/s11802-021-4436-6.
73. M. A. Mofaddel, A. Younes, and H. H. El-Sayed, Multi-objective multicast routing based on ant colony optimization in mobile ad-hoc networks, *Journal of Statistics Applications & Probability*. **10**(3), 687–694, (2021). doi: 10.18576/jsap/100308.
74. X. Wang, Low-energy secure routing protocol for wsns based on multiobjective ant colony optimization algorithm, *Journal of Sensors*. **2021**, 7633054, (2021). doi: 10.1155/2021/7633054.
75. Z. ran Dong, X. yi Bian, and S. Zhao, Ship pipe route design using improved multi-objective ant colony optimization, *Ocean Engineering*. **258**, 111789, (2022). doi: 10.1016/j.oceaneng.2022.111789.
76. J.-Y. Li, X.-Y. Deng, Z.-H. Zhan, L. Yu, K. C. Tan, K.-K. Lai, and J. Zhang, A multipopulation multiobjective ant colony system considering travel and prevention costs for vehicle routing in covid-19-like epidemics, *IEEE Transactions on Intelligent Transportation Systems*. **23**(12), 25062–25076, (2022). doi: 10.1109/TITS.2022.3180760.
77. J. Liu, Y. Dong, Z. Liu, and D. Chen, Applying ontology learning and multi-objective ant colony optimization method for focused crawling to meteorological disasters domain knowledge, *Expert Systems with Applications*. **198**, 116741, (2022). doi: 10.1016/j.eswa.2022.116741.
78. E. Saeki, S. Bao, T. Takayama, and N. Togawa, Multi-objective trip planning based on ant colony optimization utilizing trip records, *IEEE Access*. **10**, 127825–127844, (2022). doi: 10.1109/ACCESS.2022.3227431.
79. H. Sun, Y. Chen, J. Ma, Y. Wang, X. Liu, and J. Wang, Multi-objective optimal travel route recommendation for tourists by improved ant colony optimization algorithm, *Journal of Advanced Transportation*. **2022**, 6386119, (2022). doi: 10.1155/2022/6386119.
80. S. Wang, Y. Liu, Y. Qiu, Q. Zhang, F. Huo, Y. Huangfu, C. Yang, and J. Zhou, Cooperative task allocation for multi-robot systems based on multi-objective ant colony system, *IEEE Access*. **10**, 56375–56387, (2022). doi: 10.1109/ACCESS.2022.3165198.
81. L.-J. Wu, Z.-G. Chen, C.-H. Chen, Y. Li, S.-W. Jeon, J. Zhang, and Z.-H. Zhan, Real environment-aware multisource data-associated cold chain logistics scheduling: A multiple population-based multiobjective ant colony system approach, *IEEE Transactions on Intelligent Transportation Systems*. **23**(12), 23613–23627, (2022). doi: 10.1109/TITS.2022.3203629.
82. C. Zhang, D. Zhang, M. Zhang, J. Zhang, and W. Mao, A three-dimensional ant colony algorithm for multi-objective ice routing of a ship in the arctic area, *Ocean Engineering*. **266**, 113241, (2022). doi: 10.1016/j.oceaneng.2022.113241.

83. M.-M. Sun, Z.-G. Chen, Y. Jiang, Z.-H. Zhan, and J. Zhang. A Multipopulation Ant Colony System Algorithm for Multiobjective Trip Planning. In *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 3828–3834, (2023). doi: 10.1109/SMC53992.2023.10393984.
84. R. F. V. Cruz and C. I. H. Castellanos. A cooperative multi indicator-based ant colony optimization algorithm for the mogenconvrp. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, pp. 215–218, (2024). doi: 10.1145/3638530.3654416.
85. S. Wu, A. Dong, Q. Li, W. Wei, Y. Zhang, and Z. Ye, Application of ant colony optimization algorithm based on farthest point optimization and multi-objective strategy in robot path planning, *Applied Soft Computing Journal*. **167**, 112433, (2024). doi: 10.1016/j.asoc.2024.112433.
86. J. Wei, Z.-G. Chen, Y. Jiang, J. Zhang, and Z.-H. Zhan. An Ant Colony System Approach with Local Search for Multiobjective Logistics Scheduling. In *2025 International Conference on Machine Intelligence and Nature-Inspired Computing (MIND)*, pp. 54–59, (2025). doi: 10.1109/MIND67540.2025.11351820.
87. A. Asghari, M. Mojahedivaraki, and A. Barzegarinezhad, Structure-aware graph-rag for small language models: Reducing hallucinations and improving multi-step reasoning in specialized domains, *Mathematics*. **14**(14), 2509, (2026). doi: 10.3390/math14142509.
88. I. Berlianty, M. Arifin, and M. K. Insani, Ergonomic logistics optimization: Multi-objective ant colony and fuzzy logic approach for fatigue management, *International Journal of Industrial Engineering*. **31**(1), 79–97, (2026). doi: 10.23055/ijietap.2026.33.1.11375.
89. D.-T. Chu, X.-Y. Cheng, L.-Y. Bai, and H.-F. Ling, Mcach-aco: A multi-criteria adaptive hybrid ant colony optimization for last-mile delivery vehicle routing, *Sensors*. **26**(2), 401, (2026). doi: 10.3390/s26020401.
90. Z. Deng, Y. Gao, S. Han, X. Mu, G. Bai, Y. Xue, Z. Zhu, and H. Qin, Research on the imoaco path planning algorithm for rescue auvs, *Journal of Marine Science and Engineering*. **14**(1), 13, (2026). doi: 10.3390/jmse14010013.
91. Q. Mei, J. Li, and Z. Si, Wsn secure routing planning algorithm based on moaco, *Journal of Cyber Security and Mobility*. **15**(3), 683–708, (2026). doi: 10.13052/jcsm2245-1439.1537.
92. O. Sbayti and K. Housni, Moaco-olsr : Application of multi-objective ant colony optimization to optimal path search for enhanced routing algorithm olsr in vehicular networks, *Scientific African*. **32**, e03329, (2026). doi: 10.1016/j.sciaf.2026.e03329.
93. M. Gravel, W. Price, and C. Gagné, Scheduling continuous casting of aluminum using a multiple objective ant colony optimization metaheuristic, *European Journal of Operational Research*. **143**(1), 218–229, (2002). ISSN 0377-2217. doi: https://doi.org/10.1016/S0377-2217(01)00329-0.
94. P. R. McMullen and P. Tarasewich, Multi-objective assembly line balancing via a modified ant colony optimization technique, *International Journal of Production Research*. (2006). doi: 10.1080/00207540500216748.
95. L.-P. Ding, Y.-X. Feng, J.-R. Tan, and Y.-C. Gao, A new multi-objective ant colony algorithm for solving the disassembly line balancing problem, *International Journal of Advanced Manufacturing Technology*. **48**(5–8), 761–771, (2010). doi: 10.1007/s00170-009-2303-5.
96. M. Chica, O. Córdón, S. Damas, and J. Bautista, A new diversity induction mechanism for a multi-objective ant colony algorithm to solve a real-world time and space assembly line balancing problem, *Memetic Computing*. **3**(1), 15–24, (2011). doi: 10.1007/s12293-010-0035-6.
97. B. Yagmahan, Mixed-model assembly line balancing using a multi-objective ant colony optimization approach, *Expert Systems with Applications*. **38**(10), 12453–12461, (2011). doi: 10.1016/j.eswa.2011.04.026.
98. J. Rada-Vilela, M. Chica, Óscar Córdón, and S. Damas, A comparative study of multi-objective ant colony optimization algorithms for the time and space assembly line balancing problem, *Applied Soft Computing Journal*. **13**, 4370–4382, (2013). doi: 10.1016/j.asoc.2013.06.014.
99. J. M. Szemis, G. C. Dandy, and H. R. Maier, A multiobjective ant colony optimization approach for scheduling environmental flow management alternatives with application to the river murray, australia, *Water Resources Research*. **49**(10), 6393–6411, (2013). doi: 10.1002/wrcr.20518.
100. X. Wei and G. Lim, Pareto set-based ant colony optimization for multi-objective surgery scheduling problem, *The Open Cybernetics & Systemics Journal*. **8**, 1211–1218, (2014). doi: 10.2174/1874110X01408011211.
101. M. Chica, Ó. Córdón, S. Damas, and J. Bautista, Interactive preferences in multiobjective ant colony optimisation for assembly line balancing, *Soft Comput.* **19**, 2891–2903, (2015). doi: 10.1007/s00500-014-1451-1.
102. L. Zuo, L. Shu, S. Dong, C. Zhu, and T. Hara, A multi-objective optimization scheduling method based on the ant colony algorithm in cloud computing, *IEEE Access*. **3**, 2687–2699, (2015). doi: 10.1109/ACCESS.2015.2508940.
103. J. Eaton, S. Yang, and M. Gongora, Ant colony optimization for simulated dynamic multi-objective railway junction rescheduling, *IEEE Transactions on Intelligent Transportation Systems*. **18**(11), 2980–2992, (2017). doi: 10.1109/TITS.2017.2665042.
104. R.-H. Huang and T.-H. Yu, An effective ant colony optimization algorithm for multi-objective job-shop scheduling with equal-size lot-splitting, *Applied Soft Computing*.

- 57, 642–656, (2017). doi: 10.1016/j.asoc.2017.04.062.
105. B. Zhao, H. Gao, Y. Chen, W. Guo, and M. Deng, Two-generation pareto ant colony algorithm for multi-objective job shop scheduling problem with alternative process plans and unrelated parallel machines, *J Intell Manuf.* **29**, 93–108, (2018). doi: 10.1007/s10845-015-1091-z.
 106. Z.-G. Chen, Z.-H. Zhan, Y. Lin, Y.-J. Gong, T.-L. Gu, F. Zhao, H.-Q. Yuan, X. Chen, Q. Li, and J. Zhang, Multiobjective cloud workflow scheduling: A multiple populations ant colony system approach, *IEEE Transactions on Cybernetics.* **49**(8), 2912–2926, (2019). doi: 10.1109/TCYB.2018.2832640.
 107. Z. hong Jia, Y. Wang, C. Wu, Y. Yang, X. yi Zhang, and H. ping Chen, Multi-objective energy-aware batch scheduling using ant colony optimization algorithm, *Computers & Industrial Engineering.* **131**, 41–56, (2019). doi: 10.1016/j.cie.2019.03.033.
 108. A. M. Senthil Kumar and M. Venkatesan, Multi-objective task scheduling using hybrid genetic-ant colony optimization algorithm in cloud environment, *Wireless Personal Communications.* **107**, 1835–1848, (2019). doi: 10.1007/s11277-019-06360-8.
 109. S. Rubaiee and M. B. Yildirim, An energy-aware multiobjective ant colony algorithm to minimize total completion time and energy cost on a single-machine preemptive scheduling, *Computers & Industrial Engineering.* **127**, 240–252, (2019). doi: 10.1016/j.cie.2018.12.020.
 110. F. U. Zambuk, A. Y. Gital, M. Jiya, N. A. S. Gari, B. Ja’afaru, and A. Muhammad, Efficient task scheduling in cloud computing using multi-objective hybrid ant colony optimization algorithm for energy efficiency, *International Journal of Advanced Computer Science and Applications.* **12**(3), 450–456, (2021). doi: 10.14569/IJACSA.2021.0120353.
 111. W. Shao, Z. Shao, and D. Pi, An Ant Colony Optimization Behavior-Based MOEA/D for Distributed Heterogeneous Hybrid Flow Shop Scheduling Problem Under Nonidentical Time-of-Use Electricity Tariffs, *IEEE Transactions on Automation Science and Engineering.* **19**(4), 3379–3394, (2022). doi: 10.1109/TASE.2021.3119353.
 112. C. Zhang and F. Hao, Amaco for adaptive and efficient task allocation in medical environments, *Journal of Advanced Computational Intelligence and Intelligent Informatics.* **29**(3), 606–613, (2025). doi: 10.20965/jaciii.2025.p0606.
 113. K. Doerner, W. J. Gutjahr, R. F. Hartl, C. Strauss, and C. Stummer, Pareto Ant Colony Optimization: A Metaheuristic Approach to Multiobjective Portfolio Selection, *Annals of Operations Research.* **131**, 79–99, (2004). doi: 10.1023/B:ANOR.0000039513.99038.C6.
 114. S. M. Vieira, J. M. C. Sousa, and T. A. Runkler. Multi-criteria ant feature selection using fuzzy classifiers. In *Swarm Intelligence for Multi-objective Problems in Data Mining*, pp. 19–36. Springer, (2009). doi: 10.1007/978-3-642-03625-5_2.
 115. Y.-J. He and Z.-F. Ma, Optimal design of linear sensor networks for process plants: A multi-objective ant colony optimization approach, *Chemometrics and Intelligent Laboratory Systems.* **135**, 37–47, (2014). doi: 10.1016/j.chemolab.2014.03.019.
 116. S. S. Bastiani, L. Cruz-Reyes, E. Fernandez, and C. Gomez, Portfolio optimization from a set of preference ordered projects using an ant colony based multi-objective approach, *International Journal of Computational Intelligence Systems.* **8**(Supplement 2), 41–53, (2015). doi: 10.1080/18756891.2015.1129590.
 117. P.-J. Jing and H.-B. Shen, Macoed: a multi-objective ant colony optimization algorithm for snp epistasis detection in genome-wide association studies, *Bioinformatics.* **31**(5), 634–641, (2015). doi: 10.1093/bioinformatics/btu702.
 118. J. del Sagrado, I. M. del Águila, and F. J. Orellana, Multi-objective ant colony optimization for requirements selection, *Empirical Software Engineering.* **20**, 577–610, (2015). doi: 10.1007/s10664-013-9287-3.
 119. B. Khelifa and M. R. Laouar. Urban projects planning by multi-objective ant colony optimization algorithm. In *Proceedings of the 8th International Conference on Information Systems and Technologies*, pp. 1–6, (2018). doi: 10.1145/3200842.3200856.
 120. Lucky and A. S. Girsang, Multi-objective ant colony optimization for automatic social media comments summarization, *International Journal of Advanced Computer Science and Applications.* **10**(3), 400–408, (2019). doi: 10.14569/IJACSA.2019.0100352.
 121. M. I. Zulfa, R. Hartanto, A. E. Permanasari, and W. Ali, Genaco: A multi-objective cached data offloading optimization based on genetic algorithm and ant colony optimization, *PeerJ Computer Science.* **2021**, 729, (2021). doi: 10.7717/peerj-cs.729.
 122. F. A. Ghanem, M. C. Padma, and M. A. S. Almohamadi, Aco-rnn: A multi-objective ant colony optimization and recurrent neural network framework for disaster tweet summarization, *Journal of Theoretical and Applied Information Technology.* **104**(1), 439–453, (2026). doi: 10.5281/zenodo.18260204.
 123. X. Benlian and W. Zhiquan, A multi-objective-aco-based data association method for bearings-only multi-target tracking, *Communications in Nonlinear Science and Numerical Simulation.* **12**, 1360–1369, (2007). doi: 10.1016/j.cnsns.2006.02.007.
 124. Y. Y. G. W. J. C. W. Dai, Multi-objective optimization based on ant colony optimization in grid over optical burst switching networks, *Expert Systems with Applications.* **37**(2), 1769–1775, (2010). doi: 10.1016/j.eswa.2009.07.032.
 125. R. Azimi and S. Esmaeili, Multiobjective daily volt/var control in distribution systems with distributed generation using binary ant colony optimization, *Turkish Journal*

of *Electrical Engineering and Computer Sciences*. **21**(3), 613–629, (2013). doi: 10.3906/elk-1110-16.

126. P. Golding, S. Kapadia, S. Naylor, J. Schulz, H. R. Maier, U. Lall, and M. van der Velde, Framework for minimising the impact of regional shocks on global food security using multi-objective ant colony optimisation, *Environmental Modelling and Software*. **95**, 303–319, (2017). doi: 10.1016/j.envsoft.2017.06.004.
127. Y. L. Lee, J. Loo, T. C. Chuah, and A. A. El-Saleh, Multi-objective resource allocation for lte/lte-a femtocell/henb networks using ant colony optimization, *Wireless Pers Commun*. **92**, 565–586, (2017). doi: 10.1007/s11277-016-3557-5.
128. N. M. N. Pham and V. S. Le, Applying ant colony system algorithm in multi-objective resource allocation for virtual services, *Journal of Information and Telecommunication*. **1**(4), 319–333, (2017). doi: 10.1080/24751839.2017.1356159.
129. J. Zhou, C. Wang, Y. Li, P. Wang, C. Li, P. Lu, and L. Mo, A multi-objective multi-population ant colony optimization for economic emission dispatch considering power system security, *Applied Mathematical Modelling*. **45**, 684–704, (2017). doi: 10.1016/j.apm.2017.01.001.
130. Y. Qin, H. Wang, F. Zhu, and L. Zhai, A multi-objective ant colony system algorithm for virtual machine placement in traffic intense data centers, *IEEE Access*. **6**, 58912–58923, (2018). doi: 10.1109/ACCESS.2018.2875034.
131. G. R. N. Babu and R. S. Phanikumar, Multi objective task scheduling using modified ant colony optimization in cloud computing, *International Journal of Intelligent Engineering and Systems*. **11**(3), 213–220, (2018). doi: 10.22266/ijies2018.0630.26.
132. H. Xiao, Z. Hu, and K. Li, Multi-objective vm consolidation based on thresholds and ant colony system in cloud computing, *IEEE Access*. **7**, 53441–53453, (2019). doi: 10.1109/ACCESS.2019.2912722.
133. J. Shi and W.-N. Chen. Multi-objective ant colony optimization for task allocation in vehicle-based crowdsourcing. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, pp. 67–68, (2020). doi: 10.1145/3377929.3389936.
134. S.-Z. Zhou, Z.-H. Zhan, Z.-G. Chen, S. Kwong, and J. Zhang, A multi-objective ant colony system algorithm for airline crew rostering problem with fairness and satisfaction, *IEEE Transactions on Intelligent Transportation Systems*. **22**(11), 6784–6798, (2021). doi: 10.1109/TITS.2020.2994779.
135. E. Elsedimy and F. Algarni, Mots-aco: An improved ant colony optimiser for multi-objective task scheduling optimisation problem in cloud data centres, *IET Networks*. **12**(1), 4–17, (2023). doi: 10.1049/ntw.2.12033.
136. M. Abachizadeh and M. Tahani, An ant colony optimization approach to multi-objective optimal design of symmetric hybrid laminates for maximum fundamental frequency and minimum cost, *Structural and Multidisciplinary Optimization*. **37**, 367–376, (2009). doi: 10.1007/s00158-008-0235-6.
137. A. Galehdar, D. V. Thiel, A. Lewis, and M. Randall, Multiobjective optimization for small meander wire dipole antennas in a fixed area using ant colony system, *International Journal of RF and Microwave Computer-Aided Engineering*. **19**(5), 592–597, (2009). doi: 10.1002/mmce.20382.
138. L. A. Moncayo-Martínez and D. Z. Zhang, Multi-objective ant colony optimisation: A meta-heuristic approach to supply chain design, *International Journal of Production Economics*. **131**(1), 407–420, (2011). doi: 10.1016/j.ijpe.2010.11.026.
139. A. Mousa, W. F. Abd El-Wahed, and R. Rizk-Allah, A hybrid ant colony optimization approach based local search scheme for multiobjective design optimizations, *Electric Power Systems Research*. **81**(4), 1014–1023, (2011). doi: 10.1016/j.epsr.2010.12.005.
140. J. S. Angelo, H. S. Bernardino, and H. J. C. Barbosa, Ant colony approaches for multiobjective structural optimization problems with a cardinality constraint, *Advances in Engineering Software*. **80**, 101–115, (2015). doi: 10.1016/j.advengsoft.2014.09.015.
141. R. Kleinkauf, T. Houwaart, R. Backofen, and M. Mann, antarna - multi-objective inverse folding of pseudoknot rna using ant-colony optimization, *BMC Bioinformatics*. **16**, 389, (2015). doi: 10.1186/s12859-015-0815-6.
142. L. Wang, J. Shen, and J. Luo, Facilitating an ant colony algorithm for multi-objective data-intensive service provision, *Journal of Computer and System Sciences*. **81**, 734–746, (2015). doi: 10.1016/j.jcss.2014.11.017.
143. J. B. Q. Zuliani, M. W. Cohen, L. de Souza Batista, and F. G. Guimarães, Multi-objective topology optimization with ant colony optimization and genetic algorithms, *Computer-Aided Design & Applications*. **12**(6), 674–682, (2015). doi: 10.1080/16864360.2015.1033328.
144. M. Babaei and E. Sanaei, Multi-objective optimal design of braced frames using hybrid genetic and ant colony optimization, *Frontiers of Structural and Civil Engineering*. **10**(4), 472–480, (2016). doi: 10.1007/s11709-016-0368-4.
145. S. A. Khan and A. Mahmood, Fuzzy goal programming-based ant colony optimization algorithm for multi-objective topology design of distributed local area networks, *Neural Comput & Applic*. **31**, 2329–2347, (2019). doi: 10.1007/s00521-017-3191-5.
146. J. Martínez-Morales, H. Quej-Cosgaya, J. Lagunas-Jiménez, E. Palacios-Hernández, and J. Morales-Saldaña, Design optimization of multilayer perceptron neural network by ant colony optimization applied to engine emissions data, *Science China Technological Sciences*. **62**(6), 1055–1064, (2019). doi: 10.1007/s11431-017-9235-y.

147. K. Muraoka, W. Chaikittisilp, and T. Okubo, Multi-objective de novo molecular design of organic structure-directing agents for zeolites using nature-inspired ant colony optimization, *Chemical Science*. **11**, 8214–8223, (2020). doi: 10.1039/d0sc03075a.
148. H. Hu, H. Zhang, E. Shi, S. Yu, L. Huang, P. Zhou, and Y. Jin. A diffusion-based multi-objective ant colony algorithm for optimizing network topology design. In *Proceedings of the 2023 6th International Conference on Algorithms, Computing and Artificial Intelligence*, pp. 443–449, (2023). doi: 10.1145/3638884.3638945.
149. N. E. Allali, A. Zannou, O. Mahmoudi, H. Asaidi, and B. Mohamed, Dpaco: Dynamic tuning of aco with adaptive strategy for qos-aware web service composition, *Statistics, Optimization and Information Computing*. **15**, 3780–3811, (2026). doi: 10.19139/soic-2310-5070-3355.
150. N. Baskar, R. Saravanan, P. Asokan, and G. Prabhakaran, Ants Colony Algorithm Approach for Multi-Objective Optimisation of Surface Grinding Operations, *The International Journal of Advanced Manufacturing Technology*. **23**, 311–317, (2004). doi: 10.1007/s00170-002-1533-6.
151. G. N. S. Kumar, A. K. Mahendra, and G. Gouthaman, Multi-objective shape optimization using ant colony coupled computational fluid dynamics solver, *Computers & Fluids*. **46**(1), 298–305, (2011). doi: 10.1016/j.compfluid.2011.01.016.
152. E. Hajibandeh and N. Sara, Pressure zoning approach for leak detection in water distribution systems based on a multi objective ant colony optimization, *Water Resour Manage*. **32**, 2287–2300, (2018). doi: 10.1007/s11269-018-1929-1.
153. A. F. Zobaa, Mixed-integer distributed ant colony multi-objective optimization of single-tuned passive harmonic filter parameters, *IEEE Access*. **7**, 44862–44870, (2019). doi: 10.1109/ACCESS.2019.2903910.
154. N. Liu, B. Huang, and X. Pan, Using the Ant Algorithm to Derive Pareto Fronts for Multiobjective Siting of Emergency Service Facilities, *Transportation Research Record*. **1935**(1), 120–129, (2005). doi: 10.1177/0361198105193500114.
155. W. Tippachon and D. Rerkpreedapong, Multiobjective optimal placement of switches and protective devices in electric power distribution systems using ant colony optimization, *Electric Power Systems Research*. **79**(7), 1171–1178, (2009). doi: 10.1016/j.epsr.2009.02.006.
156. J. Liu and J. Liu, Applying multi-objective ant colony optimization algorithm for solving the unequal area facility layout problems, *Applied Soft Computing Journal*. **74**, 167–189, (2019). doi: 10.1016/j.asoc.2018.10.012.
157. S. Huang, Y. Zhu, K. Guo, Z. Wang, and Y. Ma, Modified Pareto Strength Ant Colony Optimization for L-Band Digital Aeronautical Communication System Cell Planning, *IEEE Internet of Things Journal*. **13**(7), 14238–14249, (2026). doi: 10.1109/JIOT.2025.3648821.
158. O. Dridia, S. Krichenb, and A. Guitounic, A multiobjective hybrid ant colony optimization approach applied to the assignment and scheduling problem, *INTERNATIONAL TRANSACTIONS IN OPERATIONAL Intl. Trans. in Op. Res*. **21**, 935–953, (2014). doi: 10.1111/itor.12071.
159. D. Li, C. Liu, and K. Li, A remanufacturing logistics network model based on improved multi-objective ant colony optimization, *Journal Européen des Systèmes Automatisés*. **52**(4), 391–395, (2019). doi: 10.18280/jesa.520409.
160. L. Pasandi, M. Hooshmand, and M. Rahbar, Modified a* algorithm integrated with ant colony optimization for multi-objective route-finding: Case study yazd, *Applied Soft Computing*. **113**, 107877, (2021). doi: 10.1016/j.asoc.2021.107877.
161. C. Mu, J. Zhang, Y. Liu, R. Qu, and T. Huang, Multi-objective ant colony optimization algorithm based on decomposition for community detection in complex networks, *Soft Computing*. **23**, 12683–12709, (2019). doi: 10.1007/s00500-019-03820-y.
162. N. S. Sani, M. Manthouri, and F. Farivar, A multi-objective ant colony optimization algorithm for community detection in complex networks, *Journal of Ambient Intelligence and Humanized Computing*. **11**, 5–21, (2020). doi: 10.1007/s12652-018-1159-7.